

CS 736

Medical Image Computing (MIC)

Statistical Shape Analysis

Suyash P. Awate

Shape Analysis

- Complaints of hip pain, shoulder pain in younger people



[en.wikipedia.org/wiki/
Lucy \(Australopithecus\)](https://en.wikipedia.org/wiki/Lucy_(Australopithecus))

Hip pain may be 'hangover from evolution'

By Smitha Mundasad
Health reporter, BBC News

🕒 27 December 2016 | **Health**

🔗 Share



SCIENCE PHOTO LIBRARY

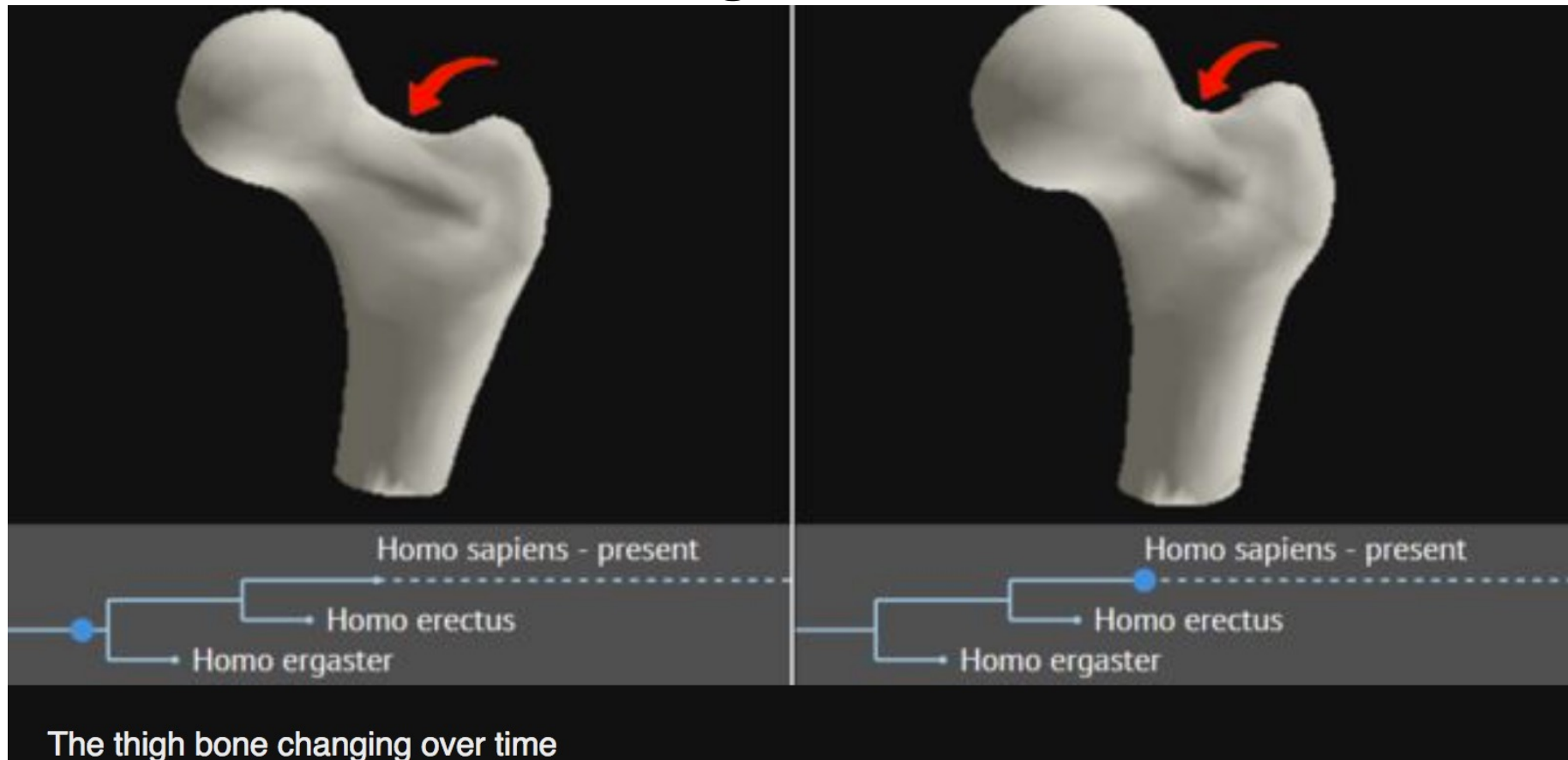
Bones from the skeleton of the 3.2m-year-old hominid Lucy

Pain in the ...

- “We wondered how on earth we have ended up with this bizarre arrangement of bones and joints that allows people to have these problems.”
- “And it struck us that the way to answer that is to look backwards through evolution.”
- Study how bones changed subtly over millennia
 - X-ray computed tomography (CT) scans of 300 ancient specimens, from different species, spanning 400 million years
 - Natural History Museum in London and Oxford
 - Smithsonian Institution, Washington

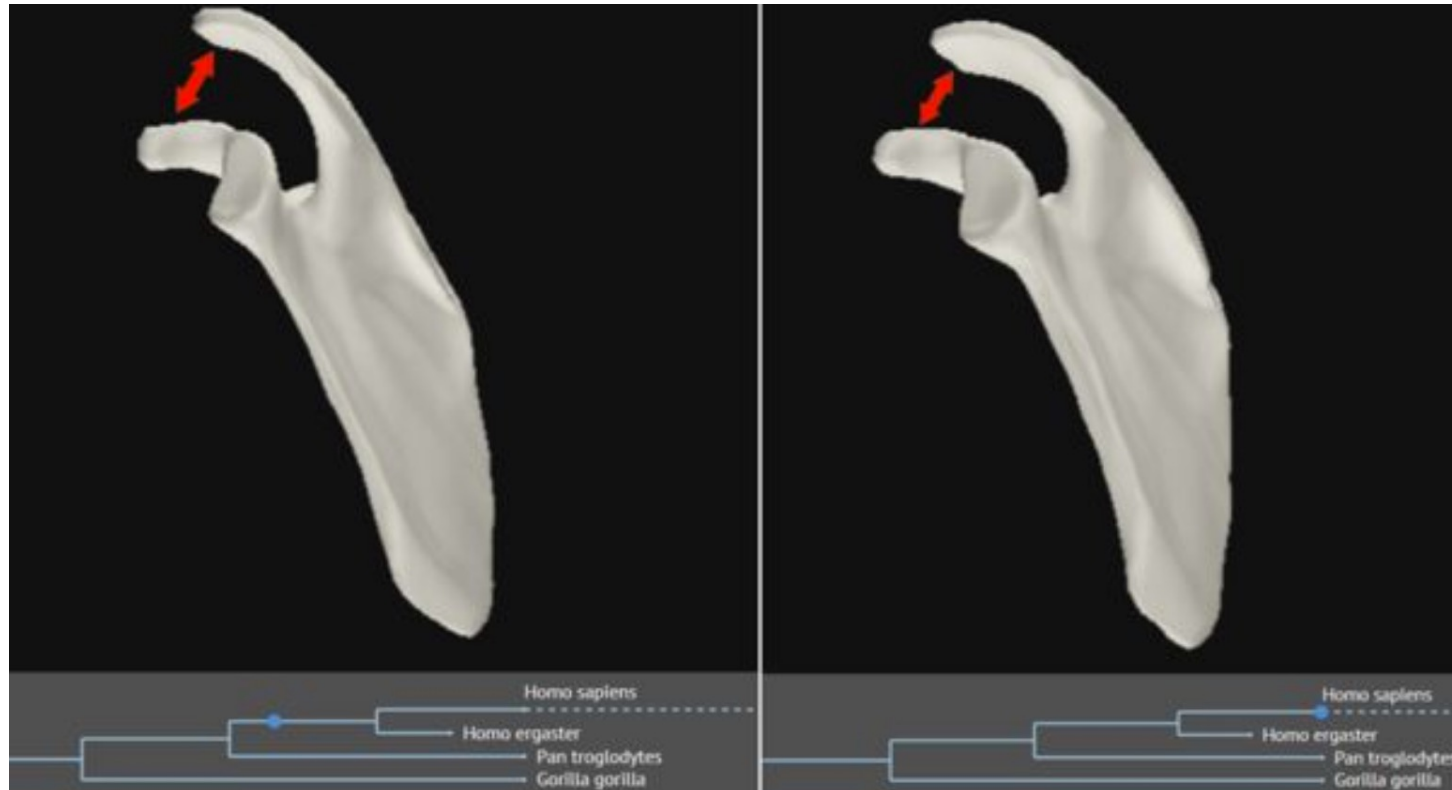
Pain in the ...

- In **hip**:
As species evolved from moving around on 4 legs to standing up on 2, **neck of thigh bone** grew broader to support extra weight
- Studies show: thicker neck of thigh bone → arthritis more likely



Pain in the ...

- In **shoulder**: a natural **gap** - which tendons and blood vessels normally pass through - got narrower over evolution
- Narrower space → more difficult for tendons to move → pain when reaching overhead



The shape of the shoulder changing over time.

Statistical Shape Analysis: References

COMPUTER VISION AND IMAGE UNDERSTANDING

Vol. 61, No. 1, January, pp. 38–59, 1995

Active Shape Models—Their Training and Application

T. F. COOTES, C. J. TAYLOR, D. H. COOPER, AND J. GRAHAM*

Department of Medical Biophysics, University of Manchester, Oxford Road, Manchester M13 9PT, England

- en.wikipedia.org/wiki/Active_shape_model

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 23, NO. 6, JUNE 2001

Active Appearance Models

tracking a single deforming
a whole class of objects.

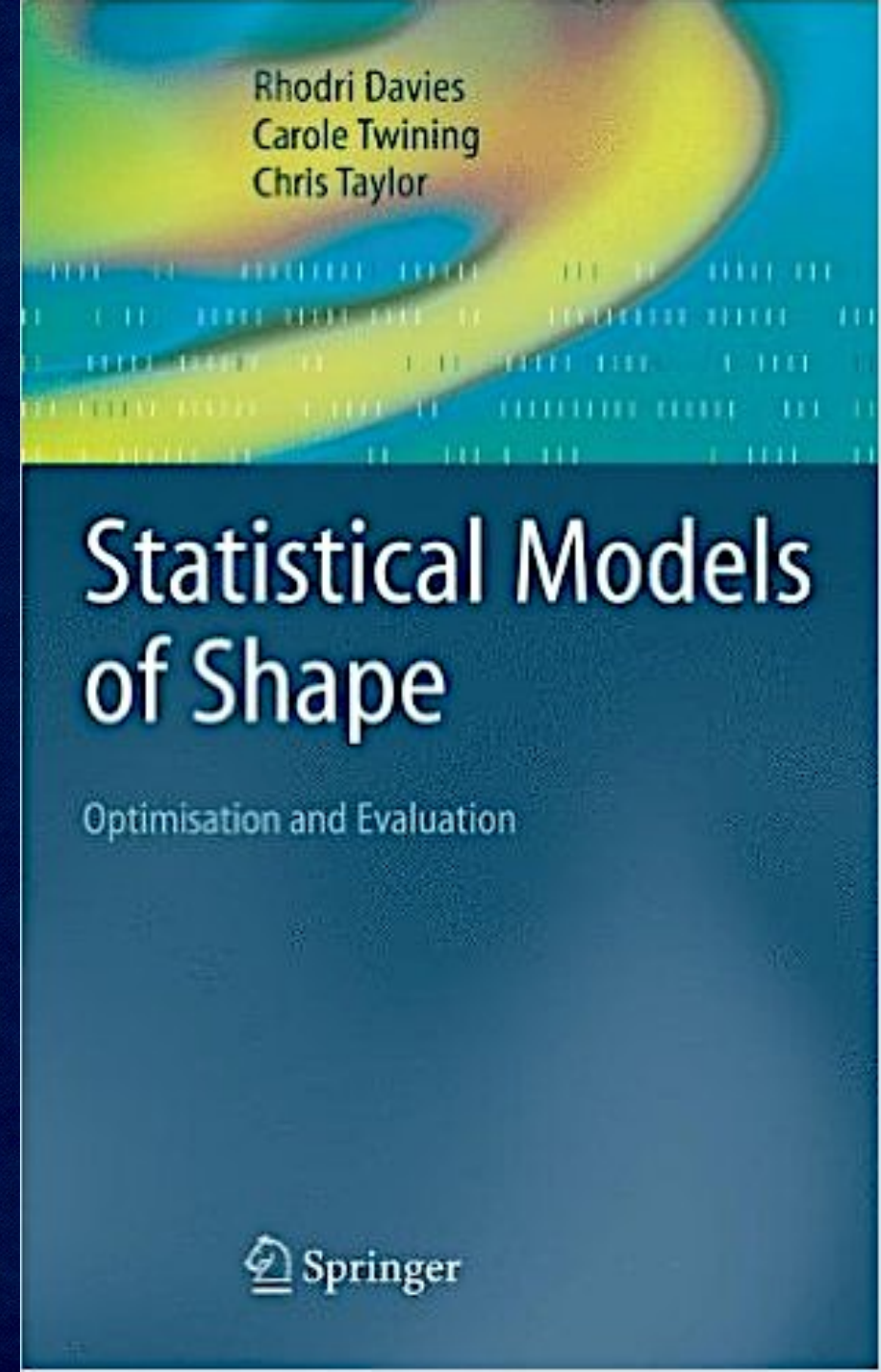
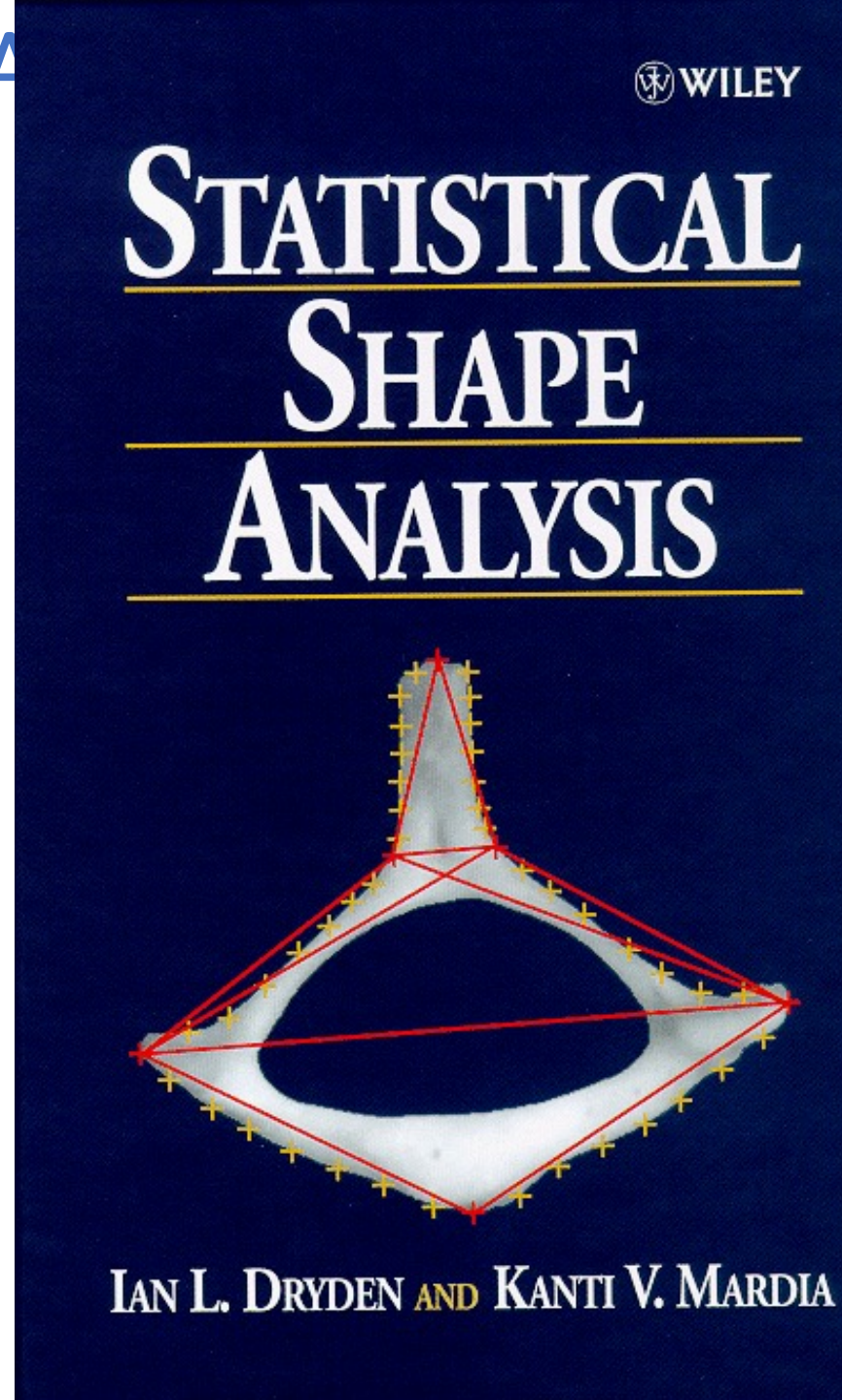
Timothy F. Cootes, Gareth J. Edwards, and
Christopher J. Taylor

- en.wikipedia.org/wiki/Active_appearance_model

2 BACKGROUND

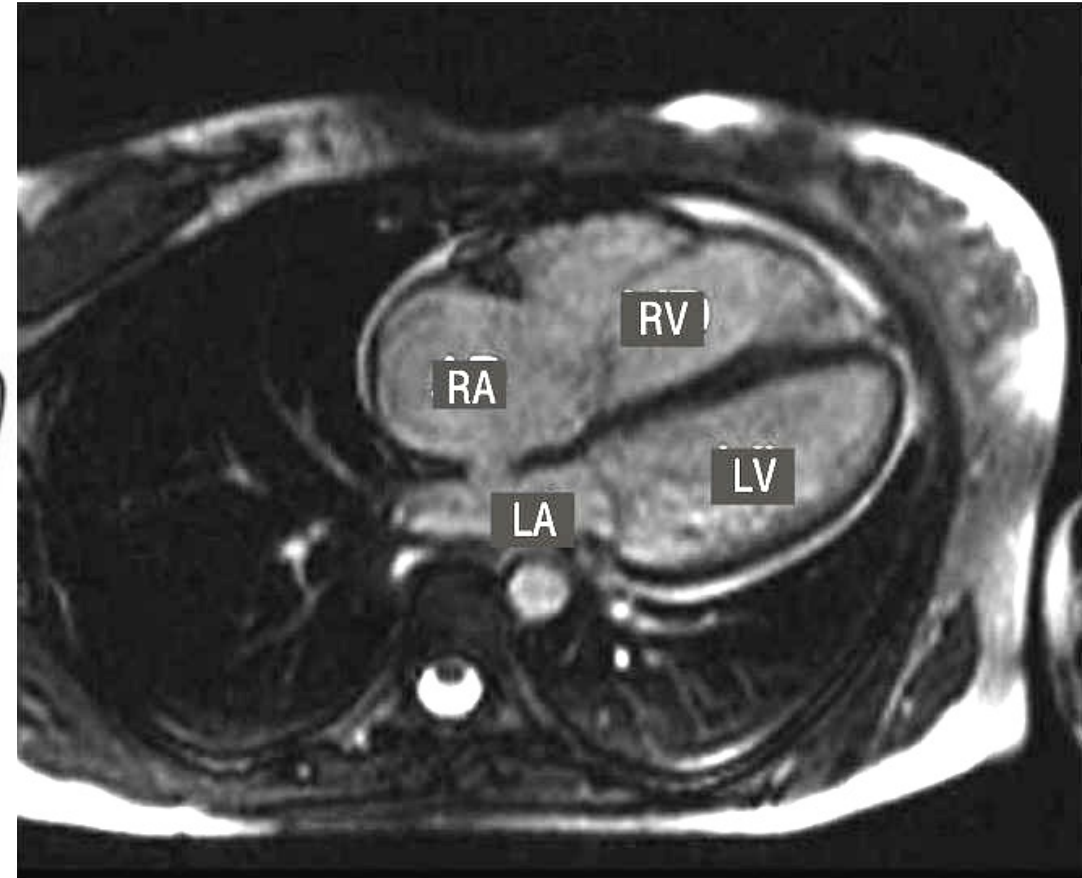
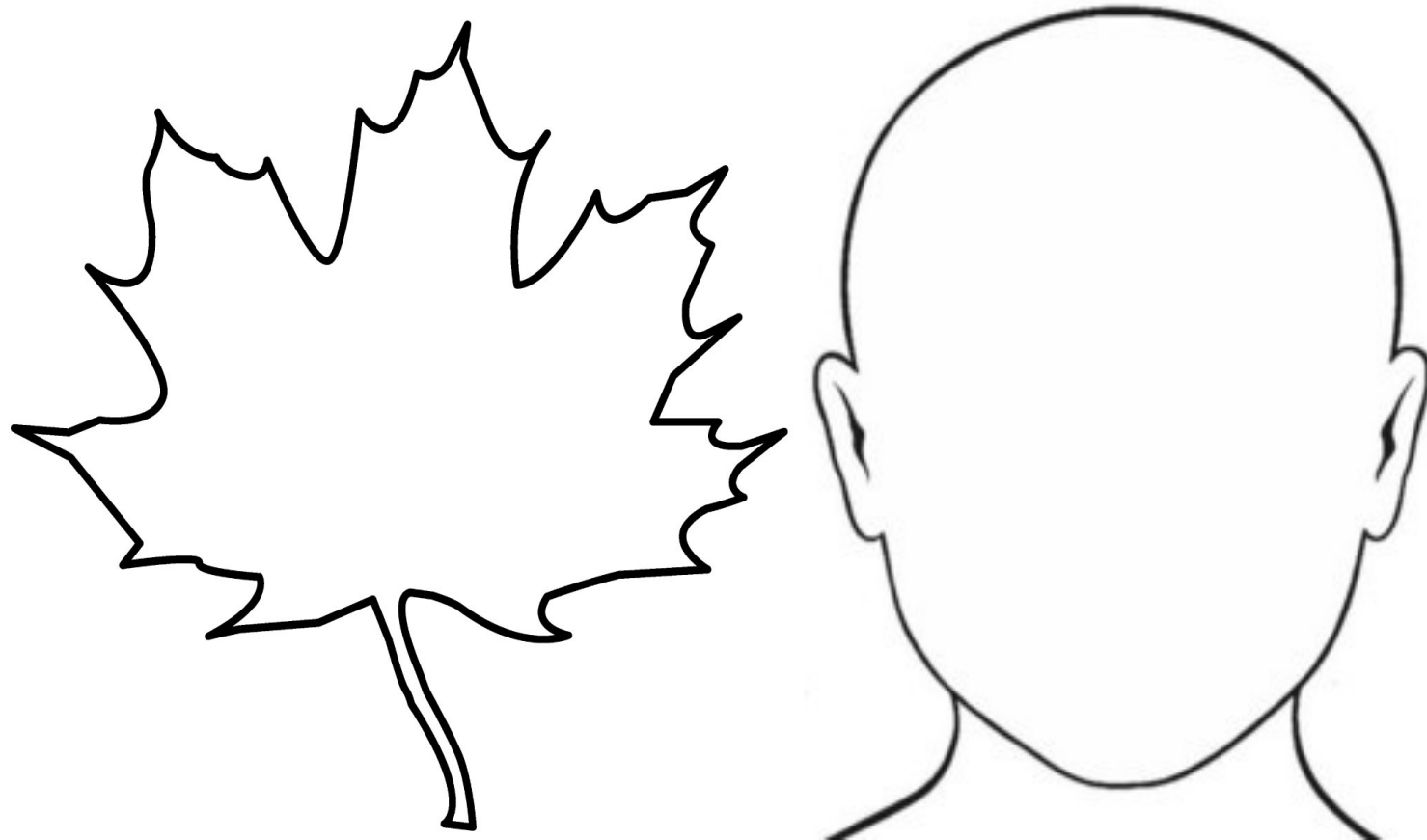
Statistical Shape Analysis

- Reference books



Shape

- What is **shape** ?
 - In plain English:
Shape = the external form, contours, or outline of someone or something
 - In geometry: there is more to it ...



Shape

- In geometry, concept of “shape” ignores location, size, pose/orientation
- Which of these outlines have the same shape ?

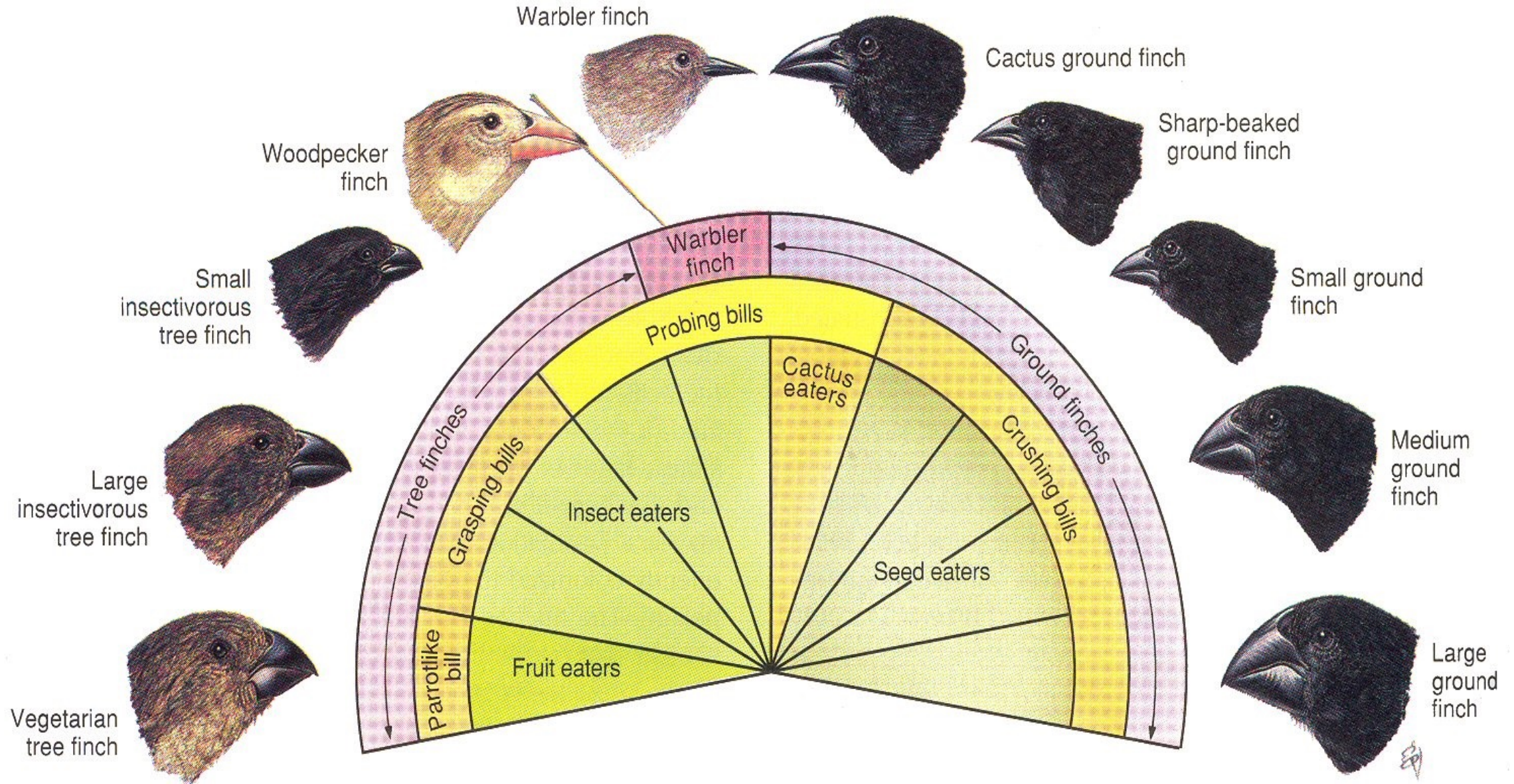


Shape

- Informally: All geometrical information that remains when location, scale, rotational effects are filtered out from (outline of) an object
 - Outlines can be in 2D (curves), 3D (surfaces), etc.
- David George Kendall pioneered statistical shape analysis
 - Kendall $\leftarrow$ Bartlett $\leftarrow$ Wishart $\leftarrow$ Karl Pearson
- More formally: In Euclidean geometry, two outlines have same (“equivalent”) shape if one can be transformed into the other by a combination of translations, rotations, and isotropic scaling
- Such transformations are called “similarity transforms”
- A shape is an equivalence class

Shape Variability

- Darwin's finches: variation of beak's shape and size

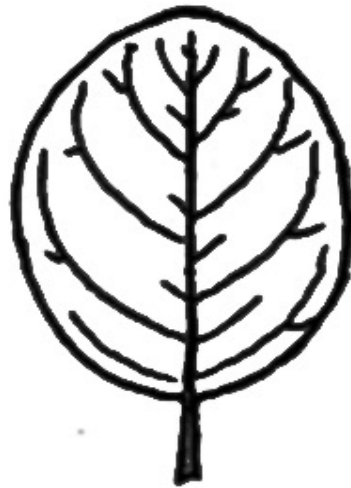


Shape Variability

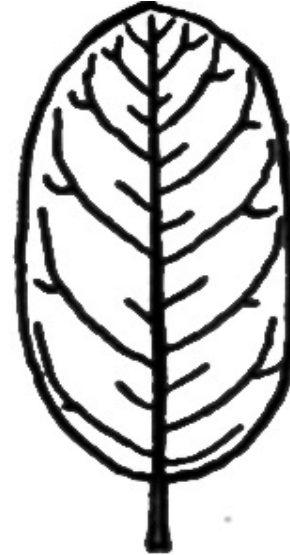
- Leaves



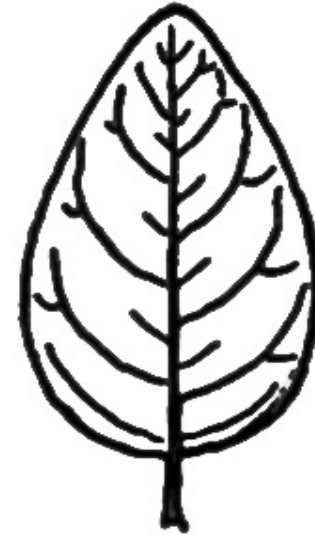
linear



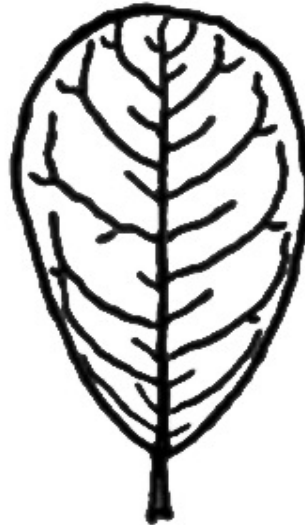
oval



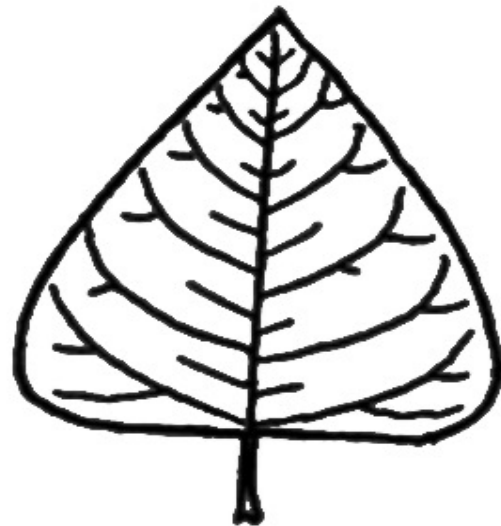
oblong



ovate



obovate



deltoid



cordate



elliptical



lanceolate

Shape Variability

- Human skull



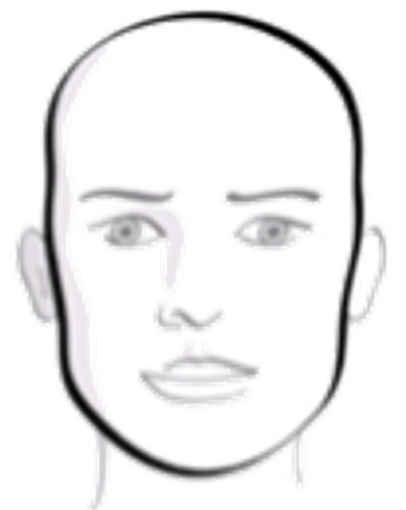
Oval



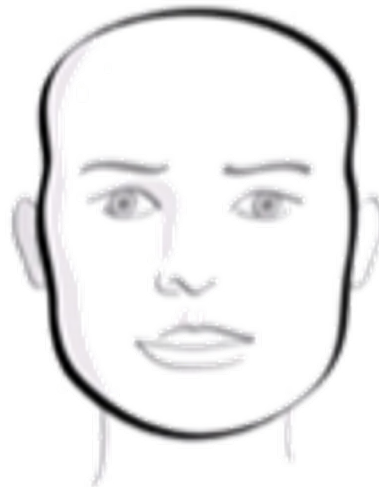
Oblong



Round



Rectangular/
Long



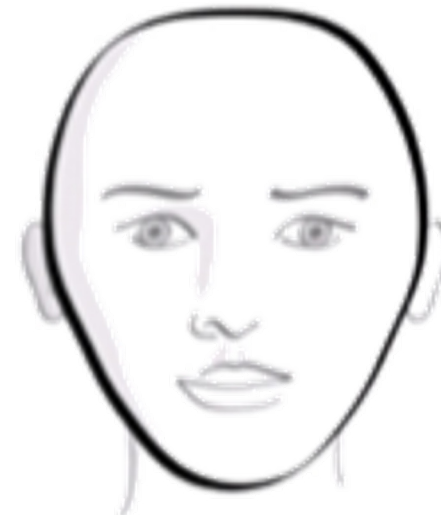
Square



Triangular



Inverted Triangle/
Heart



Diamond

Shape Variability

- Grape-leaf shapes: genetic similarities
- Method 1:
Handcrafted parametric feature space
 - Typically (very) low-dimensional
 - Given a point in feature space, cannot infer/visualize a shape well

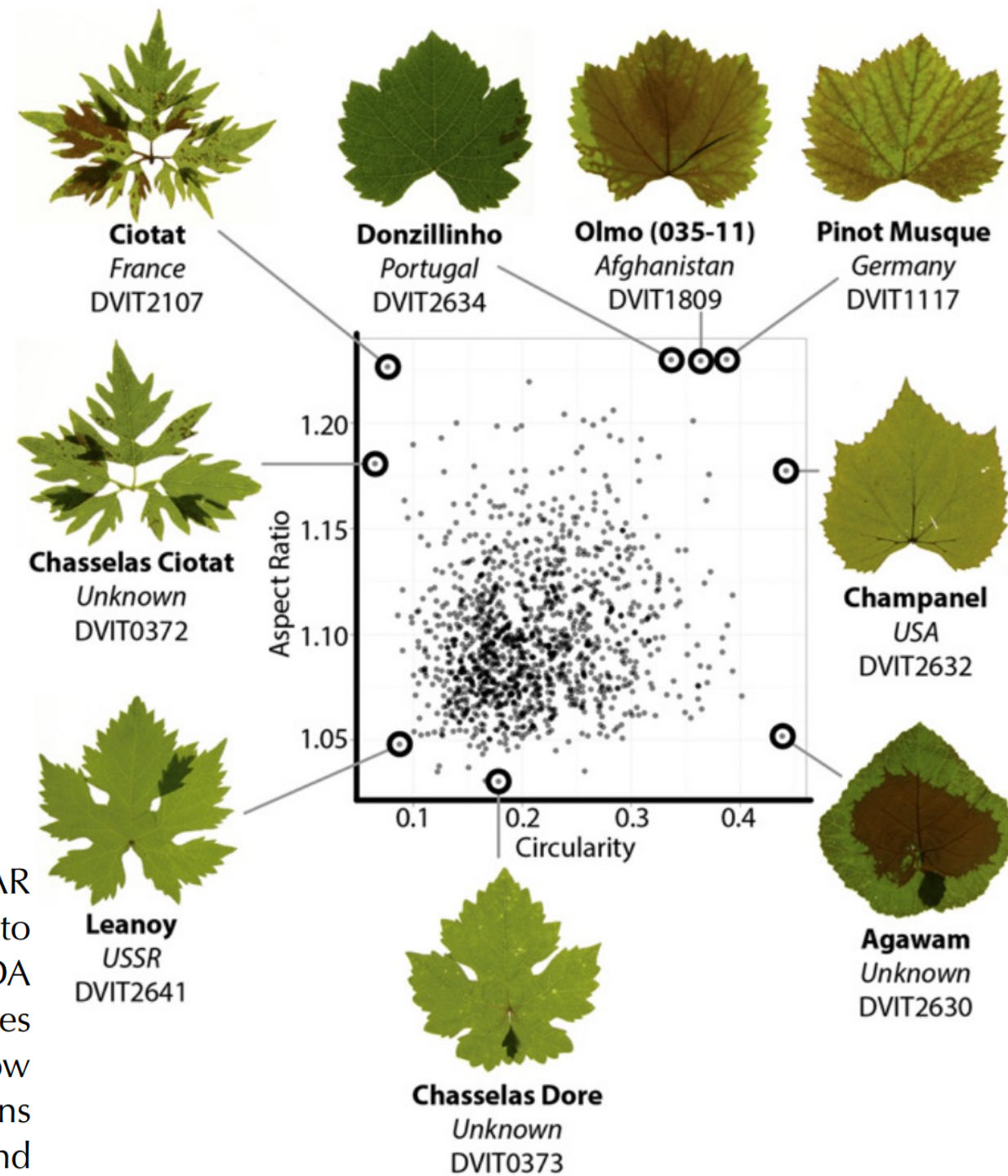
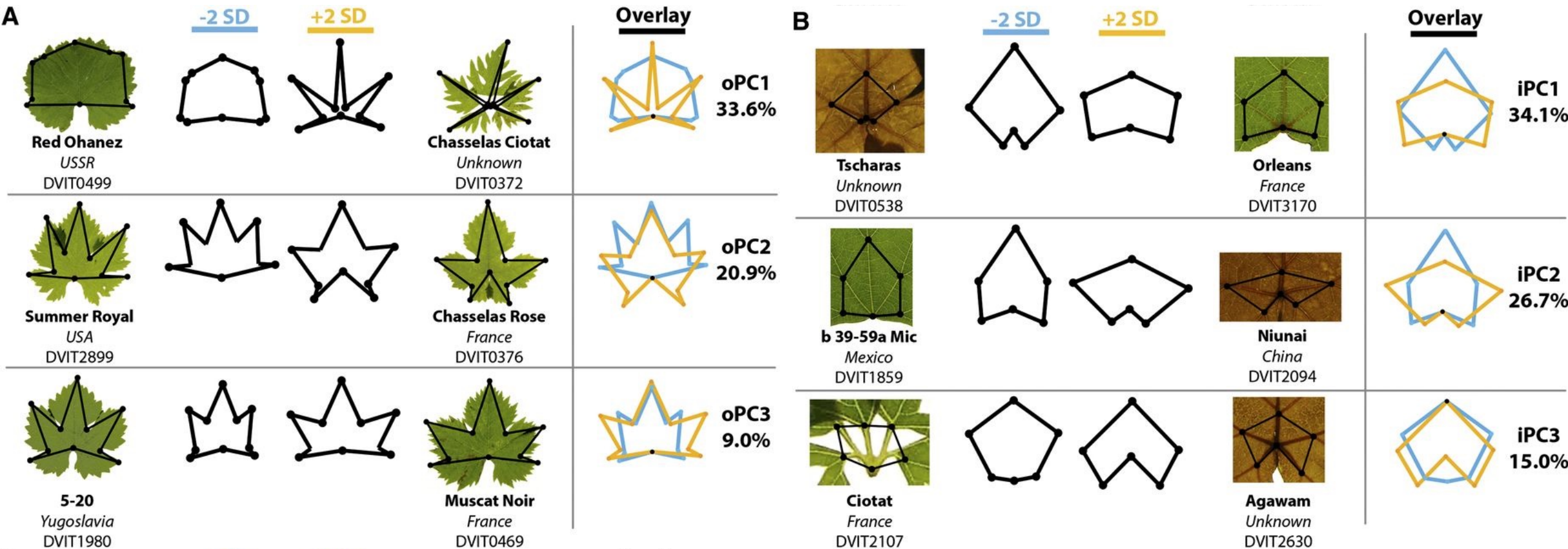


Figure 1. AR and circularity among grape accessions. Averaged AR (major/minor axis of a fitted ellipse) and circularity (the ratio of area to perimeter squared times 4π) values of 1,213 accessions in the USDA germplasm collection are shown. In this population, high AR values indicate leaves with low length-to-width ratios, and leaves with low circularity have increased lobing and serration. Leaves from accessions exhibiting extreme AR and circularity values are shown. In this and

Shape Variability

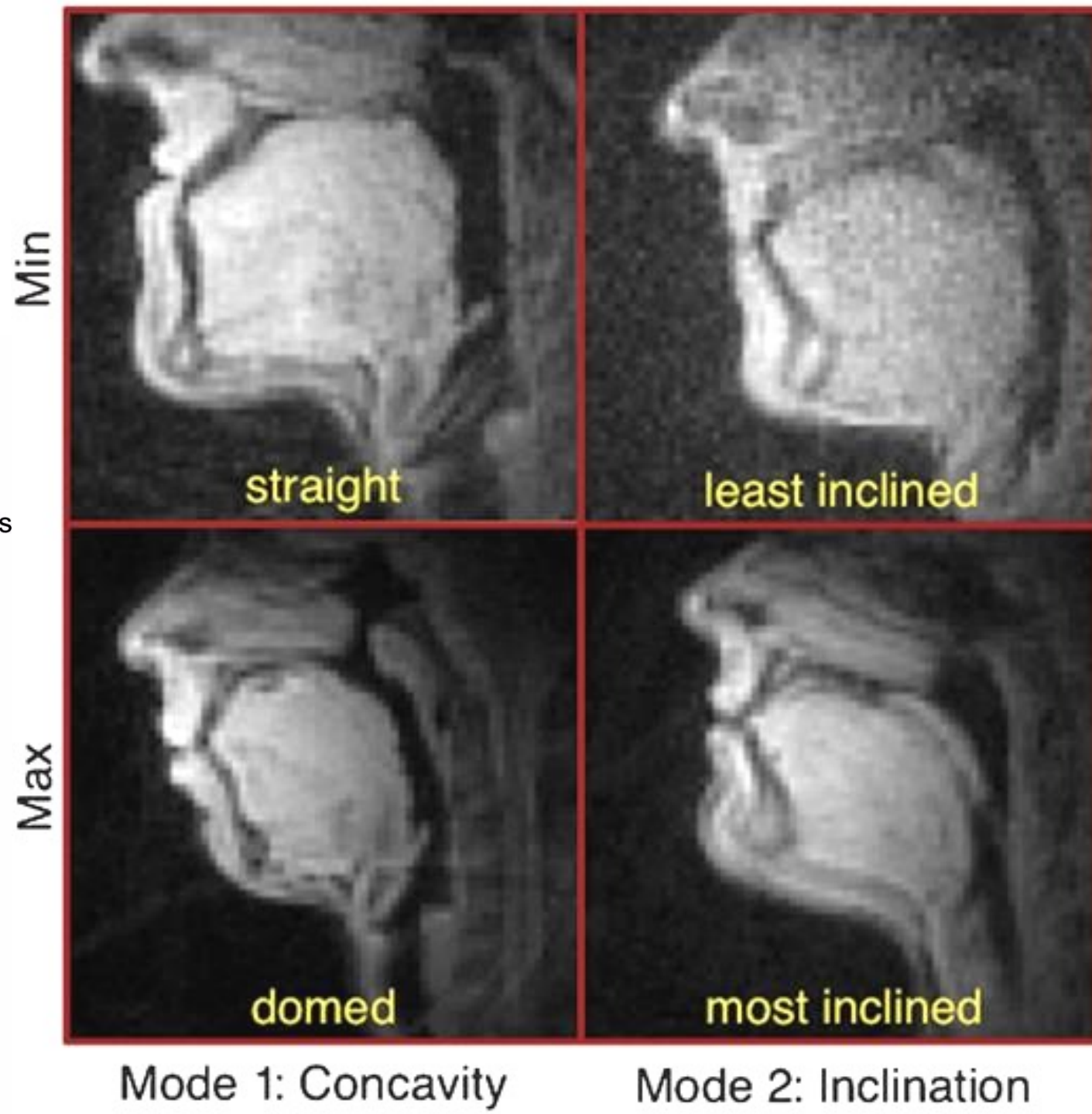
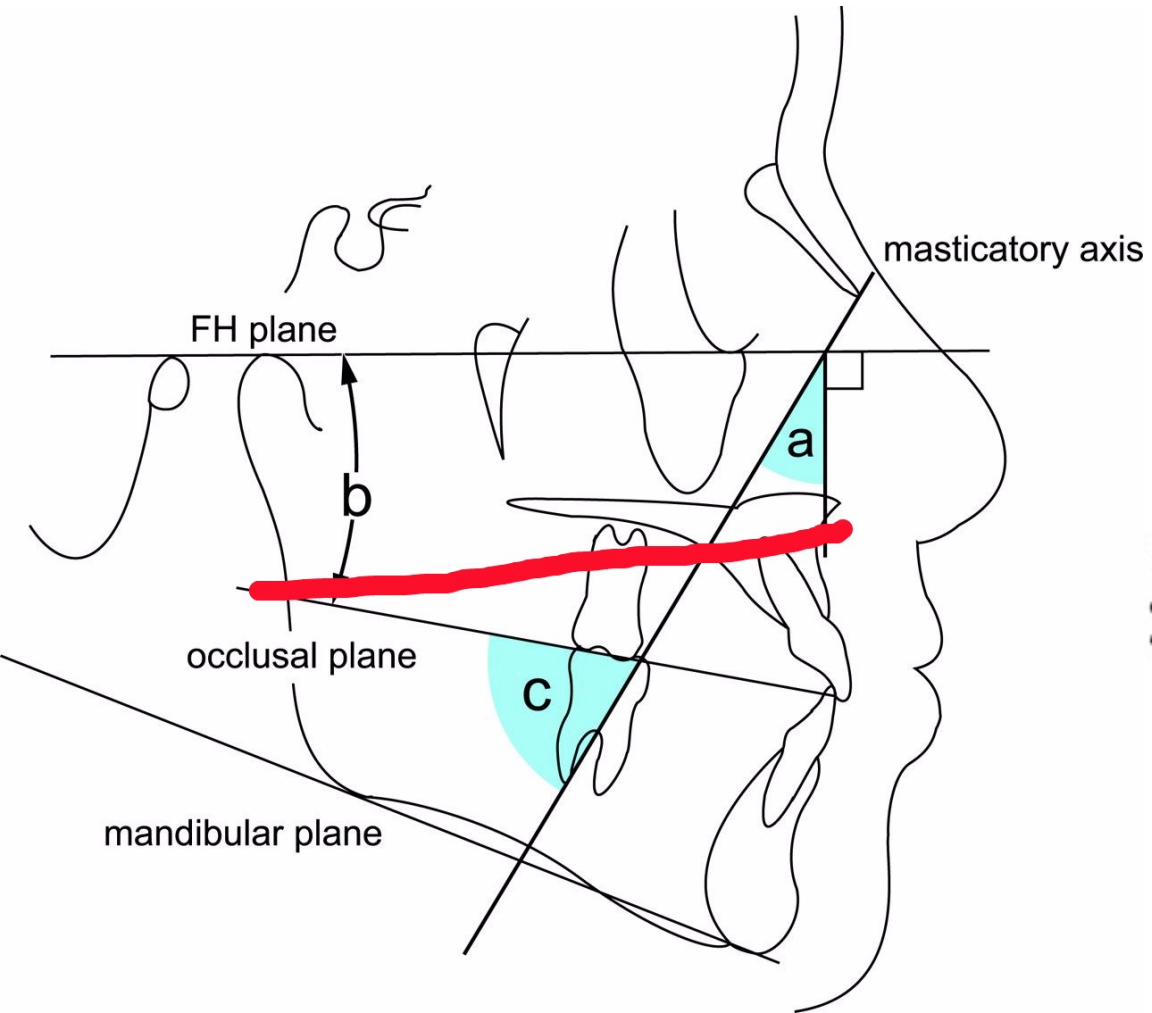
- Grape-leaf shapes: genetic similarities
- Method 2: GPA on **landmarks** (leads to rich visual representations of shape variability)
 - (A) 10 “outer” landmarks. (B) 6 “inner” landmarks.

www.plantphysiol.org/content/164/1/259



Statistical Shape Analysis

- Jaw geometry



Statistical Shape Analysis

- Normal/abnormal variation of shape of structure
 - What is **mean** (typical) shape ?
 - What is **variability** of shapes around mean shape ?

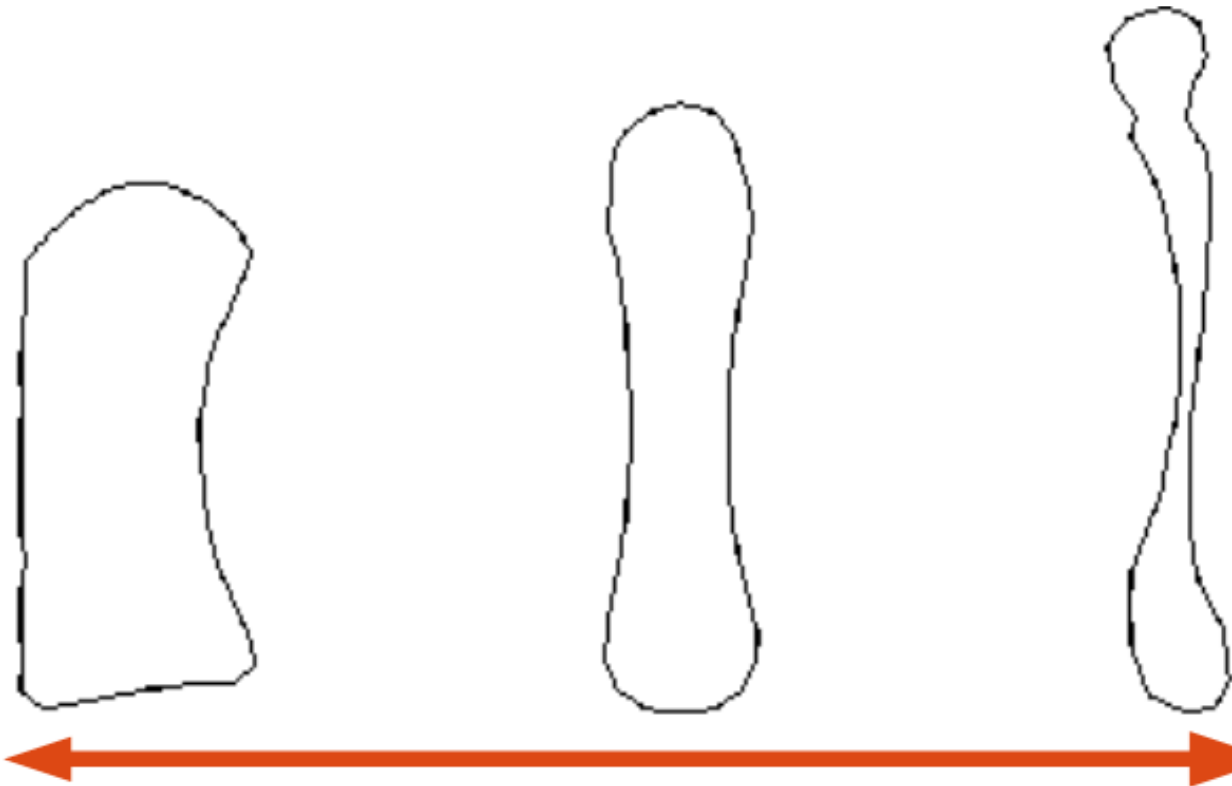
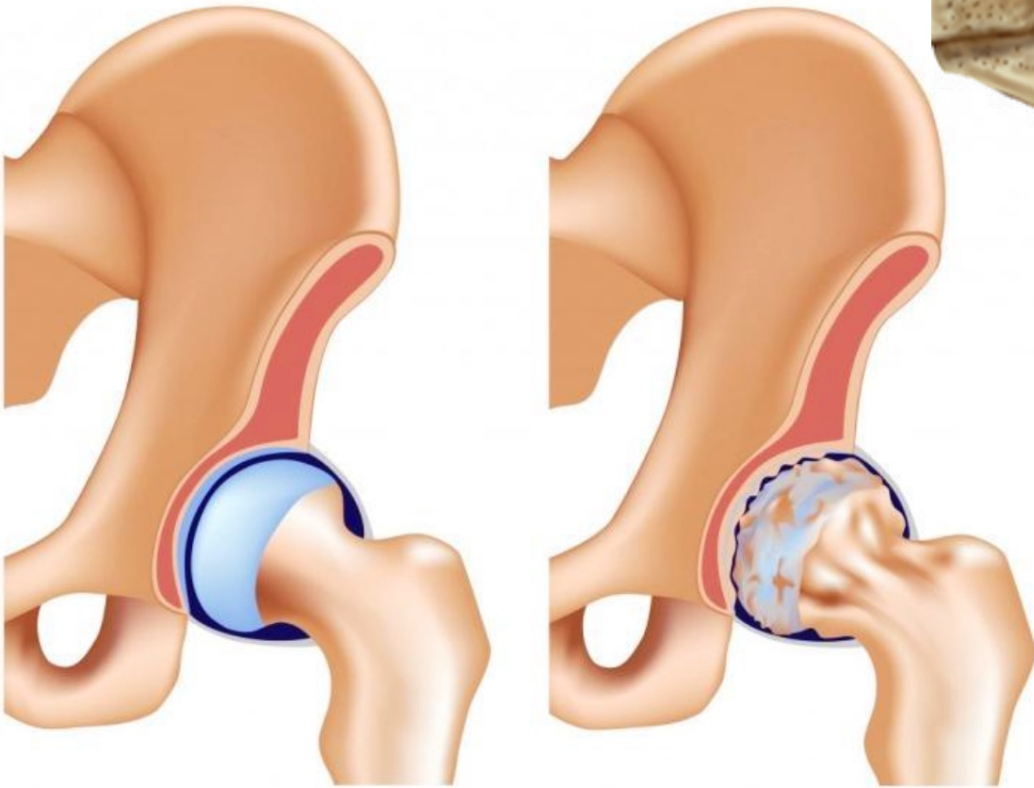


Figure 8.14 *The first mode of variation; $-2.5\lambda_1$, mean shape, $2.5\lambda_1$. Courtesy N.D. Efford, School of Computer Studies, University of Leeds.*

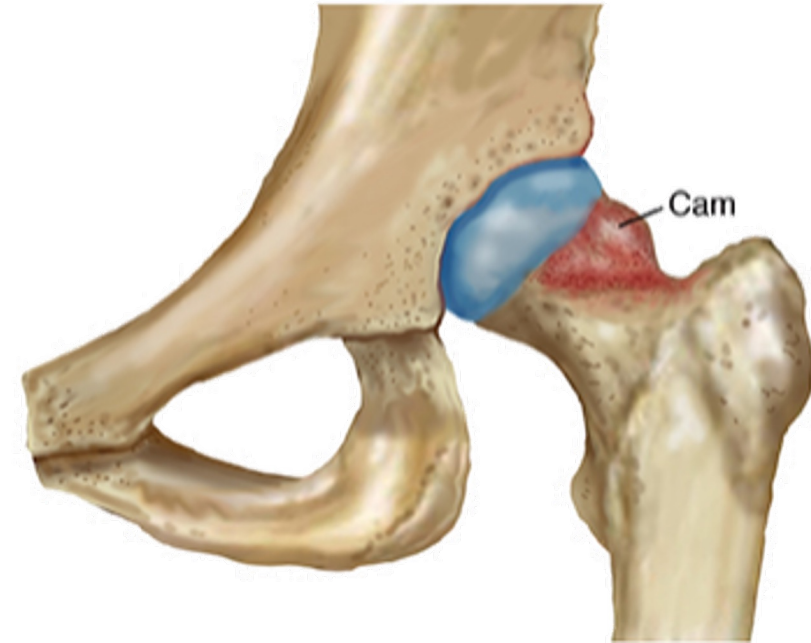
Statistical Shape Analysis

- Shape variation in normal population & diseased population



Healthy hip joint

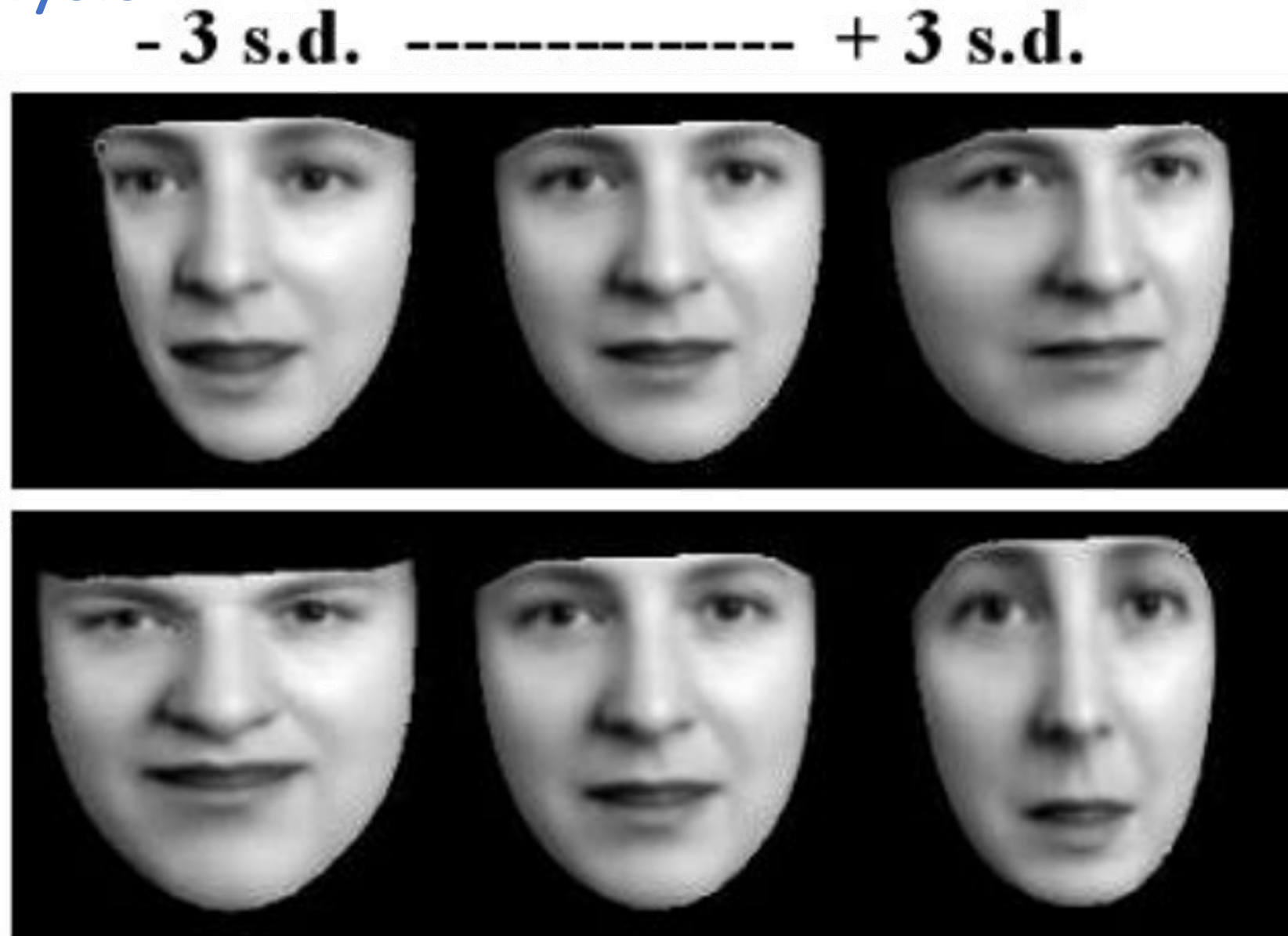
Osteoarthritis



Hip Impingement

Statistical Shape Analysis

- Face:
Geometry



First two modes of shape variation

Statistical Shape

- Face:
Geometry +
Texture

- 3 s.d. ----- + 3 s.d.



First two modes of shape variation

- 3 s.d. ----- + 3 s.d.



First two modes of gray-level variation



First four
modes of
appearance
variation

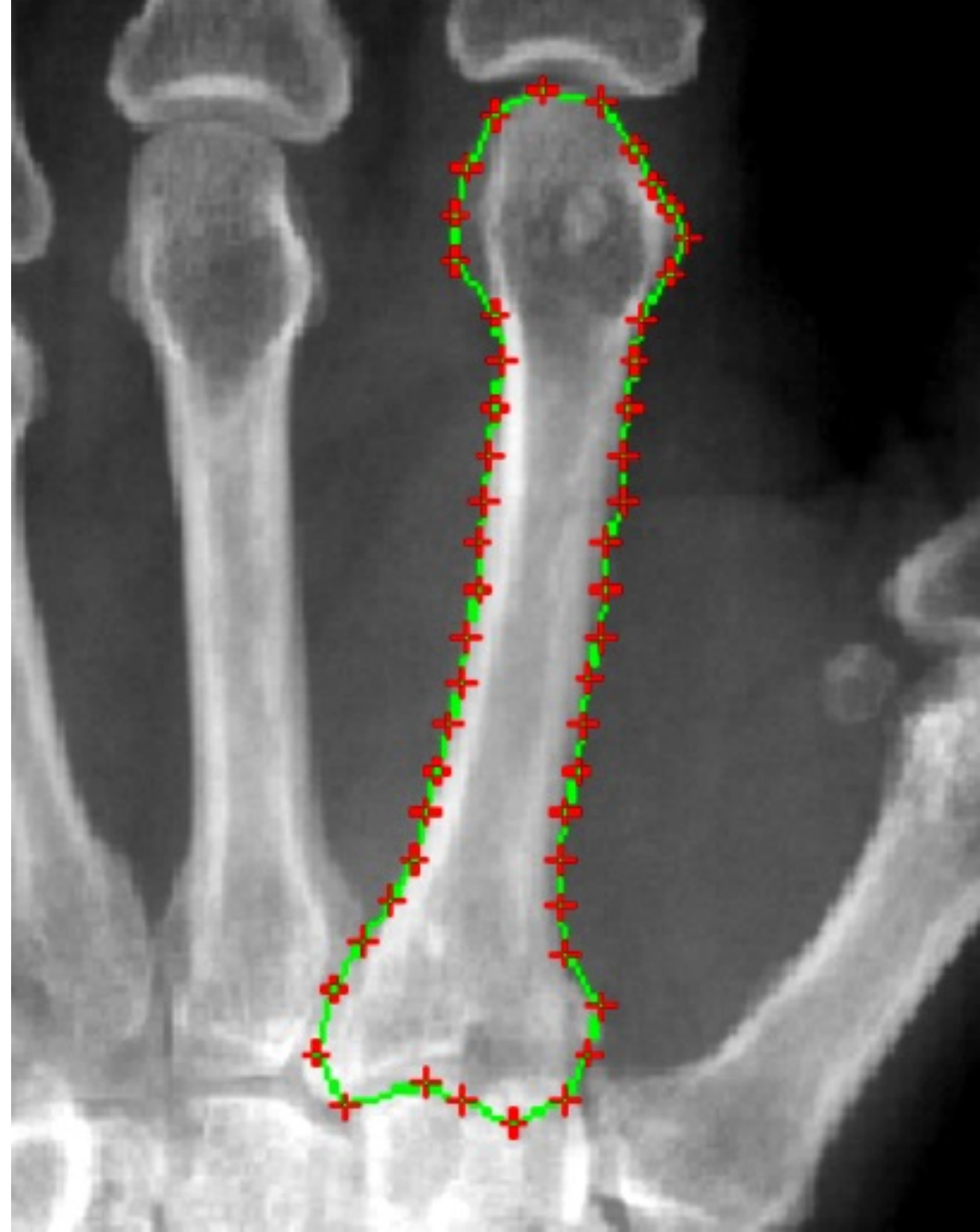


Statistical Shape Analysis

- Normal/abnormal variation of shape of structure
 - 1) To capture notion of a **typical** shape:
we need a mathematical notion of a shape “**mean**”
 - 2) To capture notion of shape **variability**,
we need a mathematical notion of shape “**covariance**”
 - Need a notion of principal **modes of variation** of shapes
 - 3) Need a way to **represent** shape that allows us to **visualize** mean and modes of variation

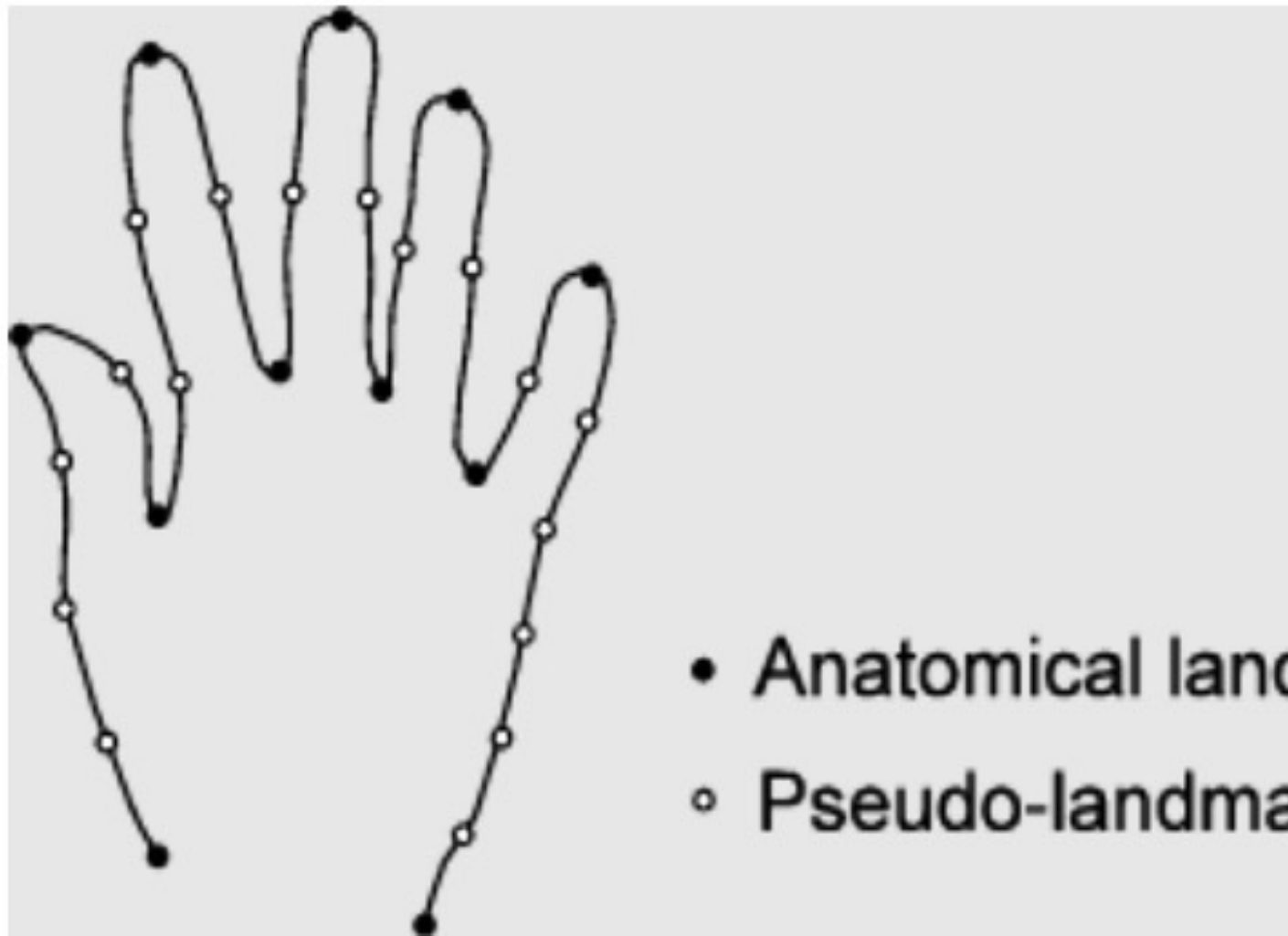
Shape Representation

- Shape represented as a pointset
- Pointset := set of points

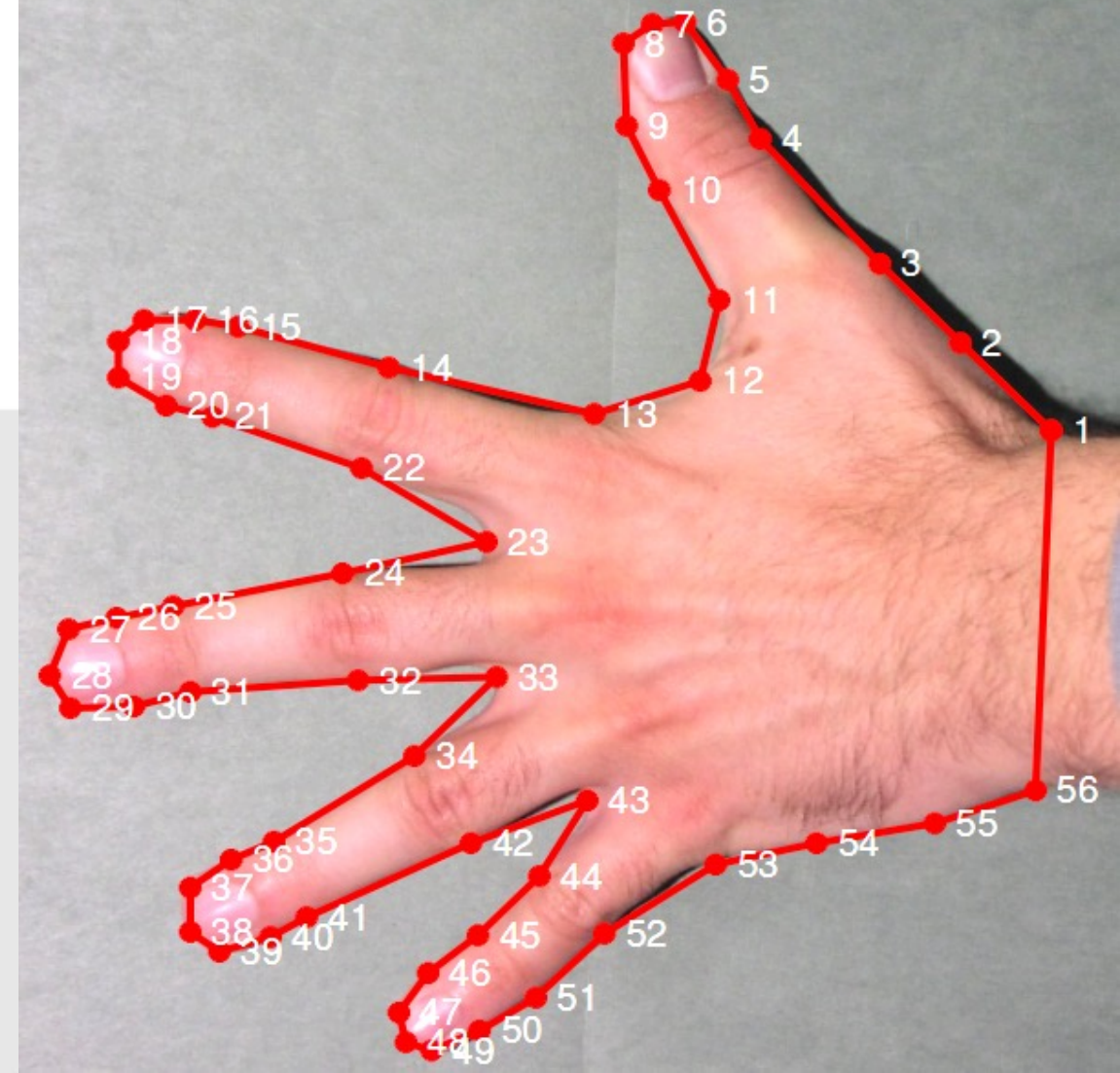


Shape Representation

- Shape represented as a pointset
- Where to place the points ? 2D case



- Anatomical landmark
- Pseudo-landmark

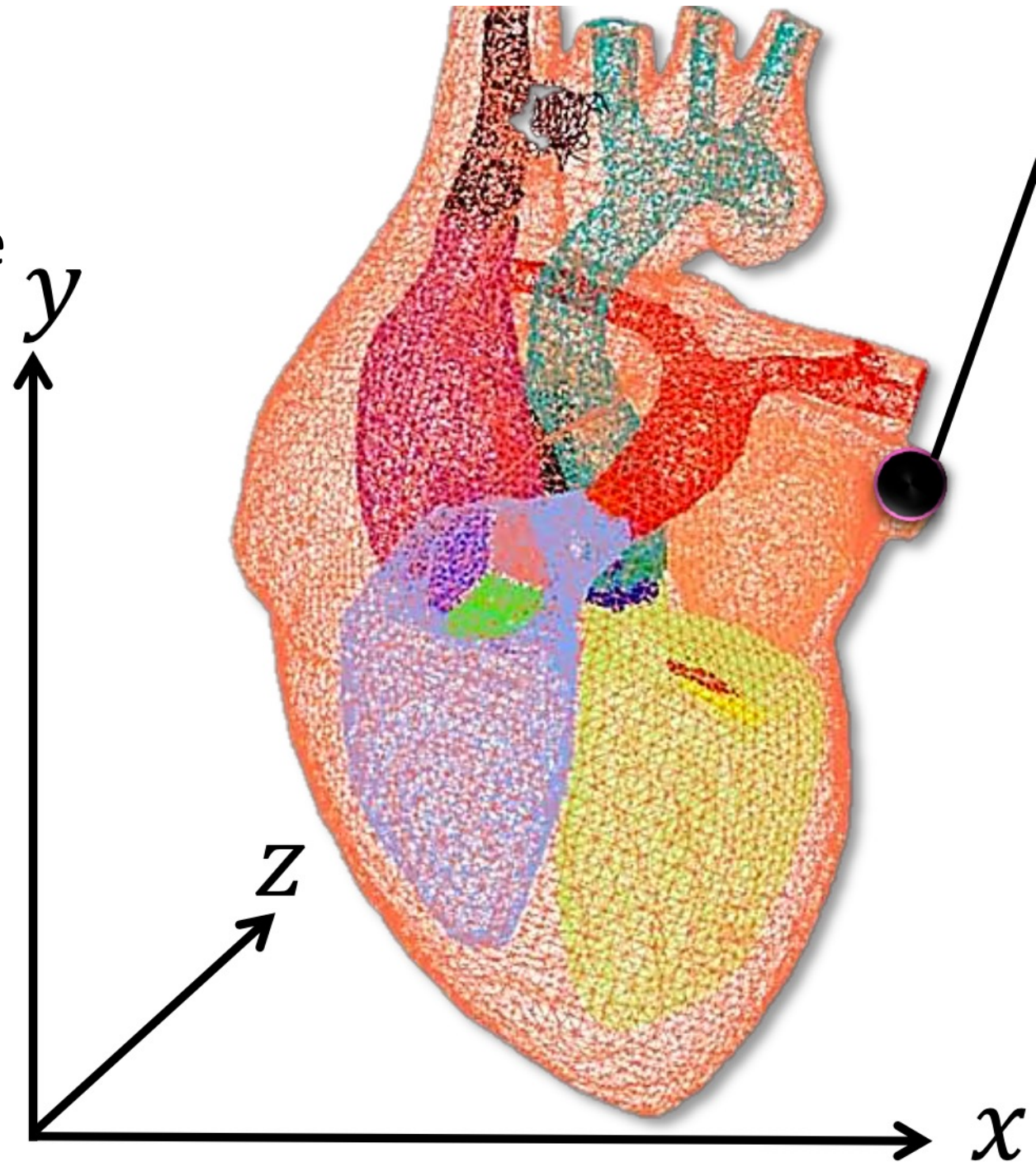


Shape Representation

- en.wikipedia.org/wiki/Landmark_point
- Some literature differentiates between anatomical landmark and mathematical landmark, but often there is a strong correlation between the two kinds
- So, we'll consider only 2 kinds of landmarks:
 - (1) pseudo and
 - (2) non-pseudo, i.e., anatomical U mathematical

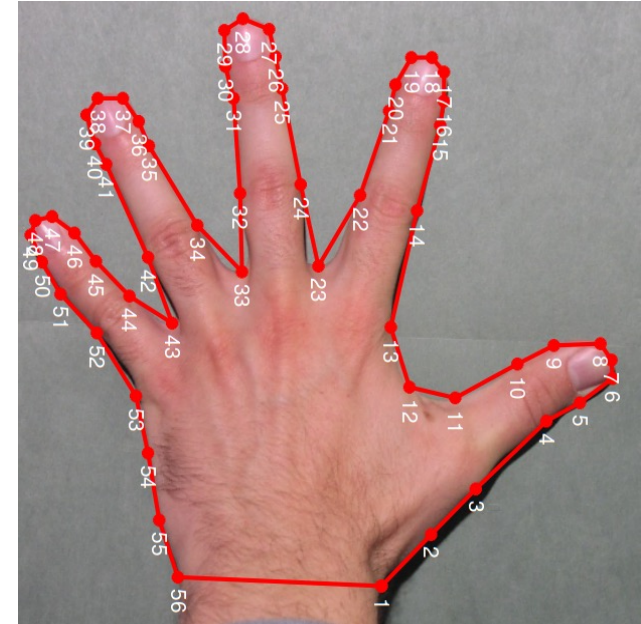
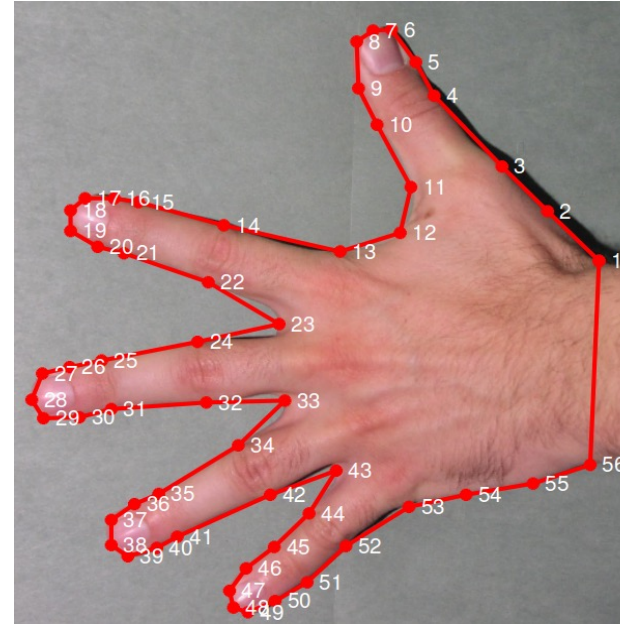
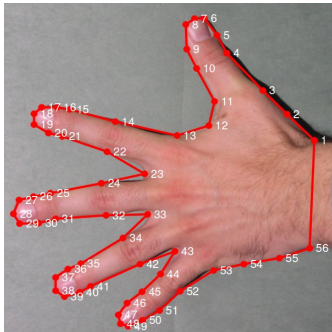
Shape Representation

- Shape represented as a pointset
- Where to place the points ? 3D case
- Doing this manually can be difficult



Shape Similarity

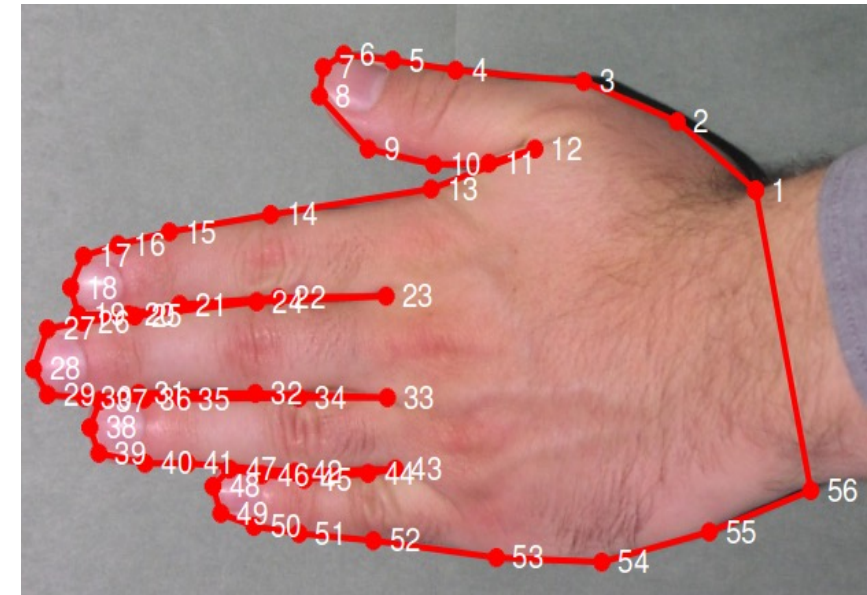
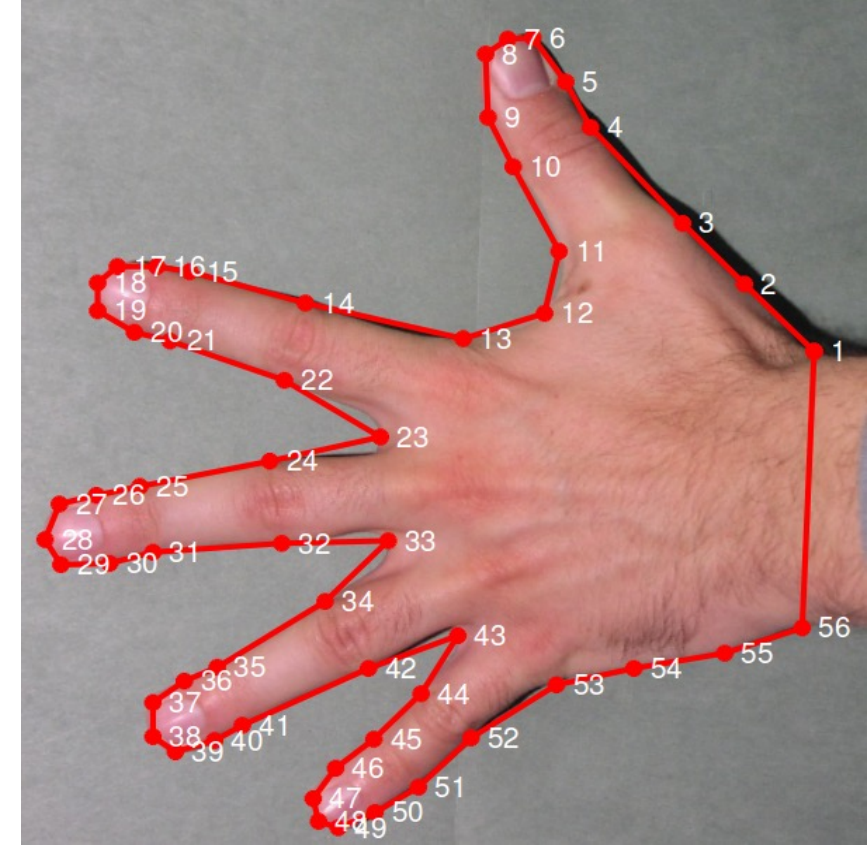
- Measuring similarity/distance between shape representations
- Assume representation = pointset
- Dissimilarity/“distance” between these shape representations equals ... ?



- Distance measure must account for following variations in pointsets:
 - Translation (coordinate-frame origin)
 - Rotation (pose)
 - Scale (size): uniform/isotropic scaling; enlarging/shrinking

Shape Similarity

- Correspondences of points across pointsets
- Our design represents each outline as a **fixed-cardinality** pointset with an **ordered** set of points placed consistently at **same landmarks** across shapes
- In this correspondence-based regime, we can choose a fixed ordering on points
→ vector of N point coordinates
- Assuming N points are in 2D physical space
 - $\text{Shape}_1 := z_1 := [z_{11}' \dots z_{1N}']'$
 $:= [x_{11} \ y_{11} \ x_{12} \ y_{12} \ \dots \ x_{1N} \ y_{1N}]'$
 - $\text{Shape}_2 := z_2 := [z_{21}' \dots z_{2N}']'$
 $:= [x_{21} \ y_{21} \ x_{22} \ y_{22} \ \dots \ x_{2N} \ y_{2N}]'$



Shape Similarity

- Pointset transformations: (special) similarity transform (in 2D)
- Translation: $T = [t_x, t_y]'$
- Scaling (identical scaling of all coordinates; NO shear): $s > 0$
- Rotation: $M_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
 - Orthogonal matrix of determinant +1
 - Rotation (counter clockwise) about origin by angle theta to +X axis
 - Reflection disallowed (hence, “special”): $M_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$
 - Reflection about line, through origin, at angle $\theta/2$ to +X axis. Matrix has determinant -1.
- Let point $a = [x_1, y_1]'$
- So, similarity transformation is: $b \leftarrow s M_\theta a + T$ where point $b = [x_2, y_2]'$

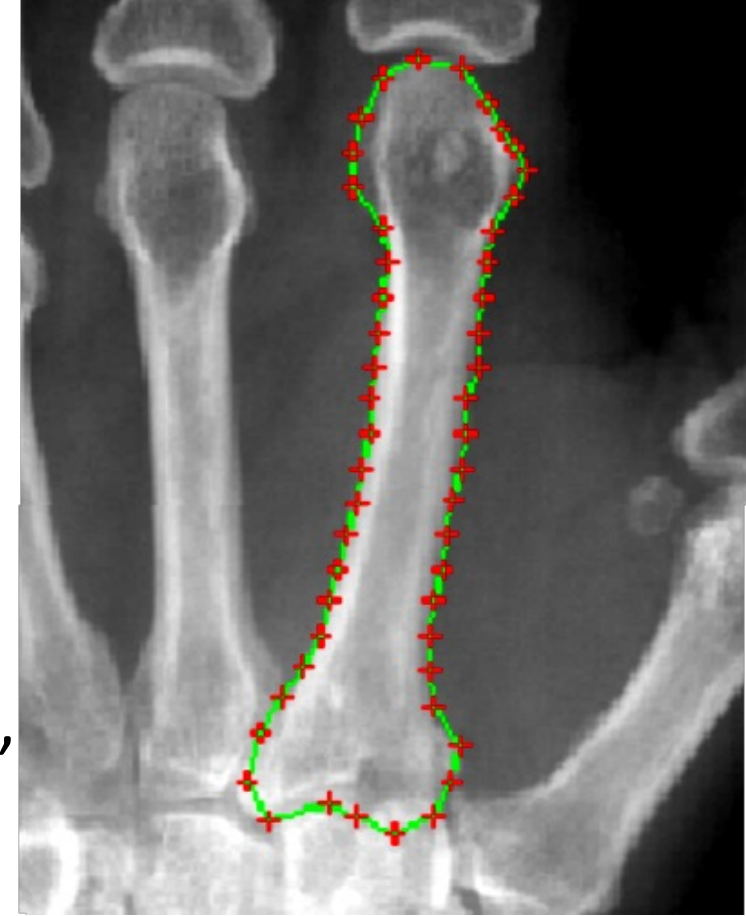
Shape Similarity

- Shape is an equivalence class of pointsets, where equivalence is determined by similarity transforms
- Are these shapes the same ?



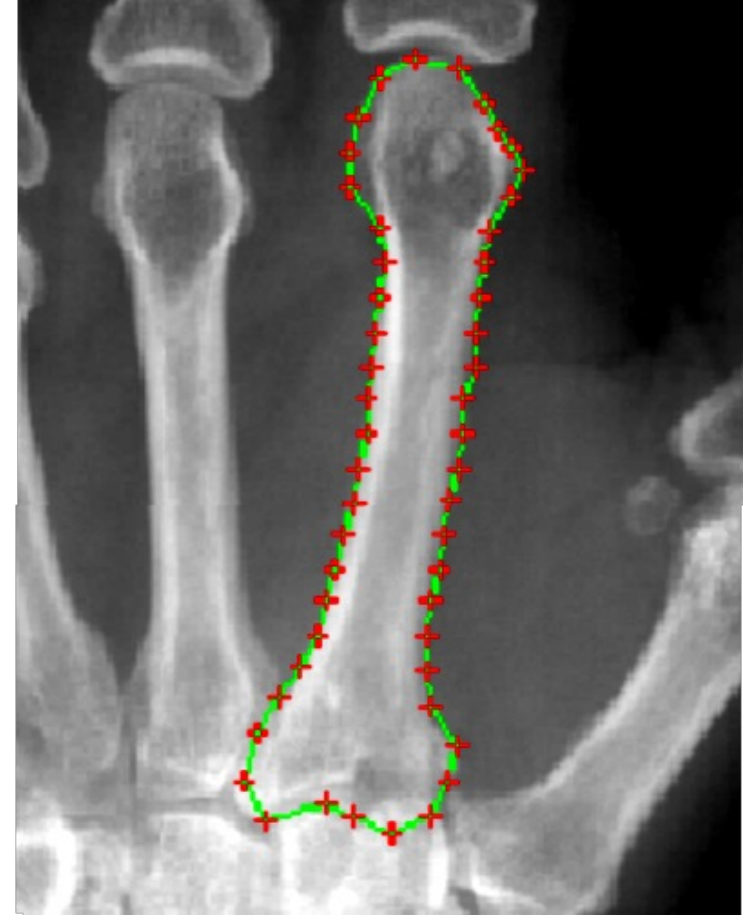
Shape Similarity

- **Shape space**: Shape in $\mathbb{R}^D$ represented by pointset $\{z_n \in \mathbb{R}^D\}_{n=1}^N$, a vector of length DN
- **Degrees of freedom (DOF)**
 - [en.wikipedia.org/wiki/Degrees_of_freedom_\(physics_and_chemistry\)](https://en.wikipedia.org/wiki/Degrees_of_freedom_(physics_and_chemistry))
 - “Minimum number of independent coordinates that can specify the position of the system completely”
- What are number of degrees of freedom associated with shape space, given invariance to translation, isotropic scaling, and rotation ?
 - Transformations of translation, scaling, and rotation are explicitly defined in physical D -dimensional space (not ND -dimensional space)



Shape Similarity

- **Shape space**: Shape in $\mathbb{R}^D$ represented by pointset $\{z_n \in \mathbb{R}^D\}_{n=1}^N$, a vector of length DN
- **Degrees of freedom (DOF) $\ll DN$**
 - 1) Boundary **regularity** reduces DOF
 - 2) If **location** (in $\mathbb{R}^D$) of shape (represented by pointset) doesn't matter, then (without loss of generality) we can specify pointset **centroid to be at origin**, i.e., $\sum_n z_n = \underline{0}$
 - Translation-invariance **reduces D DOF**: translation representation needs D DOF
 - Puts all vectorized pointsets on **same hyperplane in DN -dimensional space**



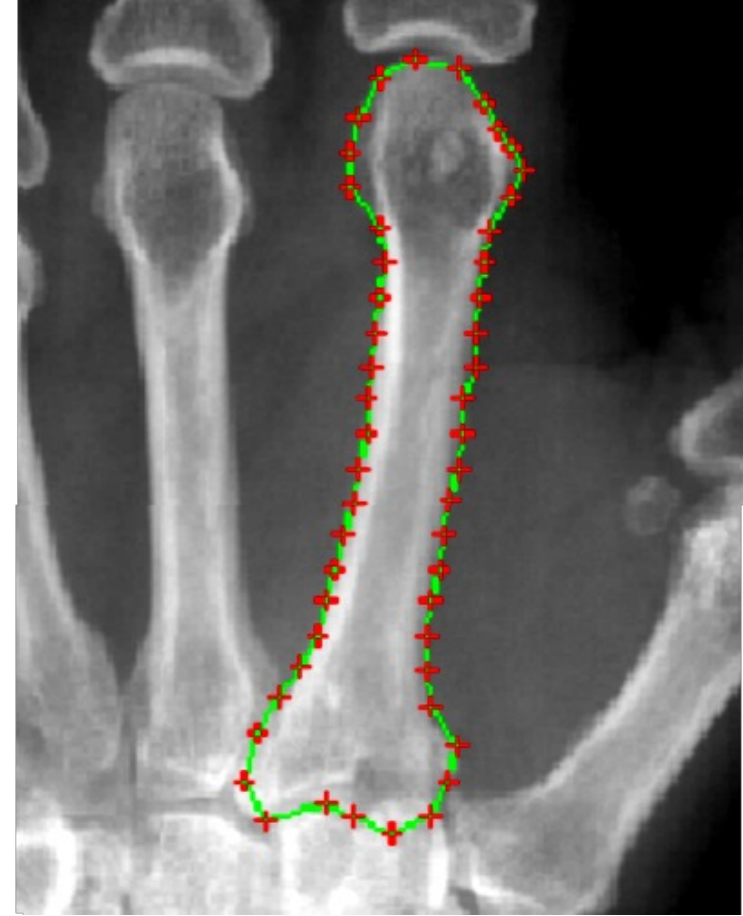
Shape Similarity

- **Shape space**: Shape in $\mathbb{R}^D$ represented by pointset $\{z_n \in \mathbb{R}^D\}_{n=1}^N$, a vector of length DN
- **Degrees of freedom (DOF) $\ll$ DN**

3) If **scale** (in $\mathbb{R}^D$)
of shape (represented by pointset)
doesn't matter,
then (without loss of generality)
we can specify pointset

“size”/norm to be fixed, i.e., $\sum_n (\|z_n\|_2)^2 = 1$

- Puts vectorized pointsets on **same hypersphere** in DN-dimensional space
- Scale-invariance **reduces 1 DOF**: Scale representation needs 1 DOF



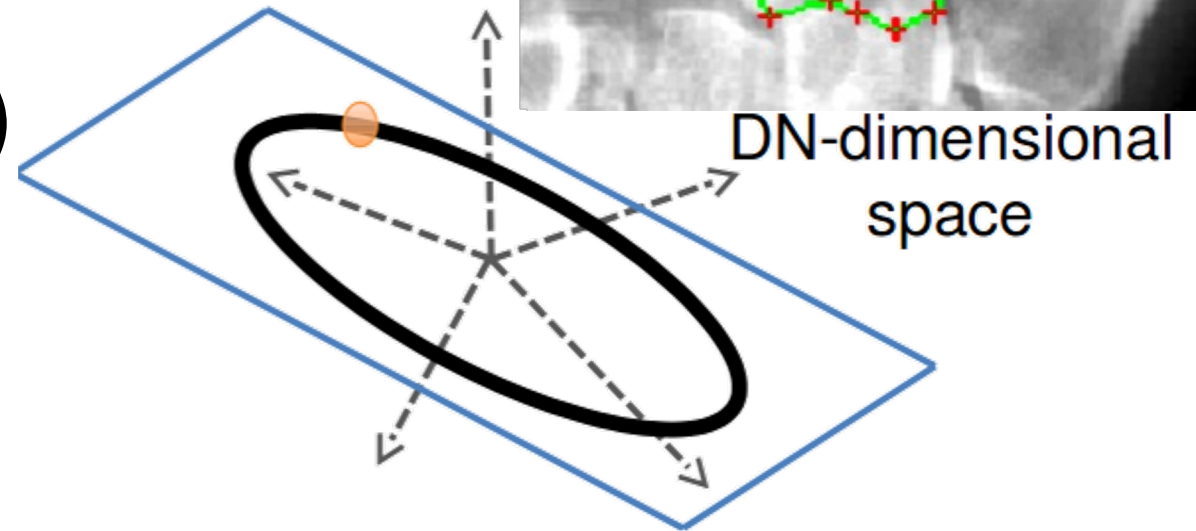
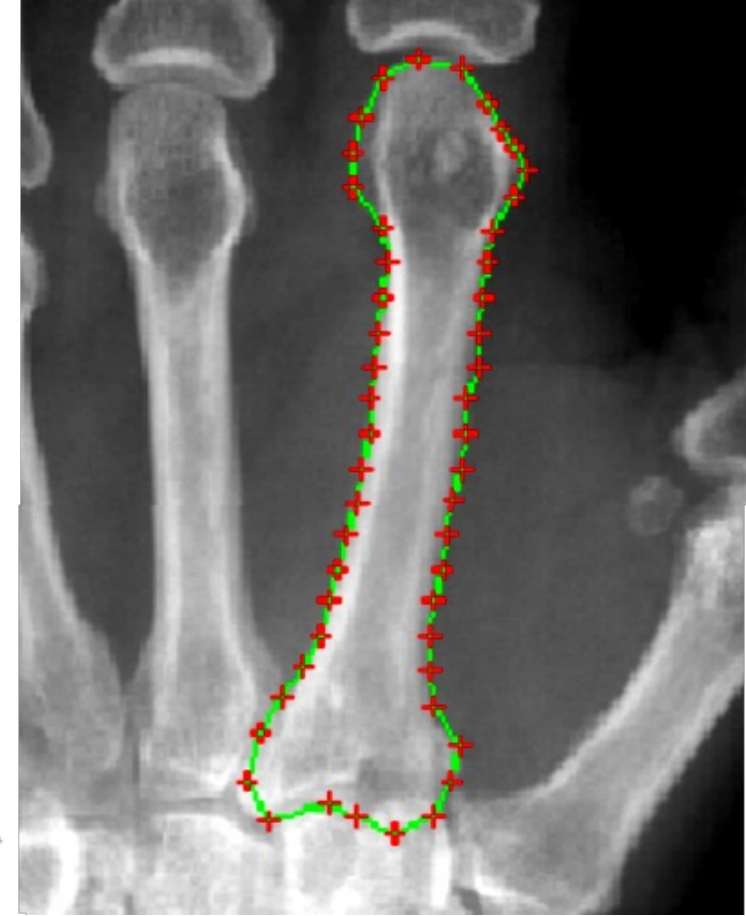
Shape Similarity

- **Shape space**: Shape in $\mathbb{R}^D$ represented by pointset $\{z_n \in \mathbb{R}^D\}_{n=1}^N$, vector of length DN
- **Degrees of freedom (DOF) $\ll$ DN**
- **Preshape space** = intersection of hyperplane and hypersphere resulting from factoring out translation & scale

4) If **rotation** (in $\mathbb{R}^D$) of shape (represented by pointset) doesn't matter, then (without loss of generality) we can **fix pointset rotation**

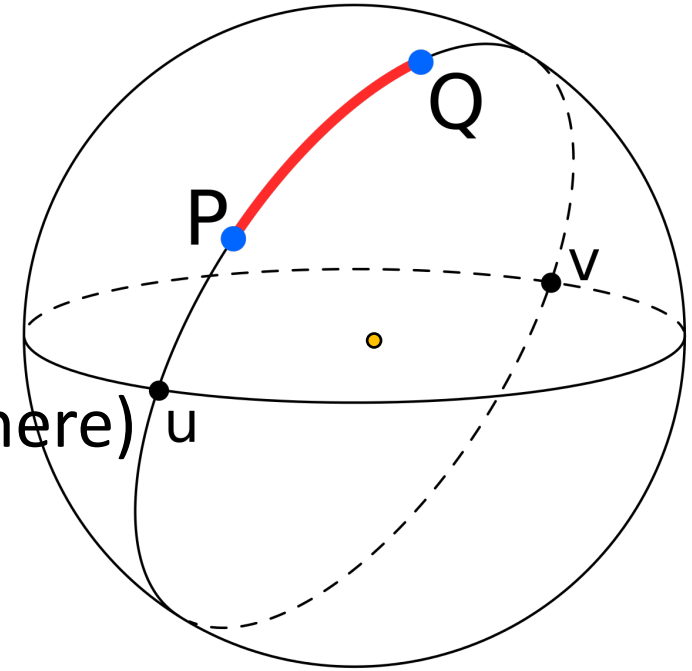
- Rotation reduces more DOF:

DOF in representation of $D \times D$ orthogonal matrix: $D \times D - (D + {}^D C_2) = D(D-1)/2$



Shape Similarity

- **Shape distance** (**squared**) between pointsets z_1 & z_2 in **preshape** shape
$$:= \Delta^2(z_1, z_2) := \min_{\theta} [\text{distance}(z_1, \text{Rotate}(z_2; \theta))]^2$$
 - Rotation, in (physical) D-dimensional space, is explicitly optimized
 - Shape distance $\Delta(z_1, z_2)$ measured, modulo rotation, in preshape space
- Two options for designing *distance* measure:
 - 1) Geodesic distance on hypersphere:
$$\text{distance}(z_1, z_2^*) := \text{GeodesicDistance}(z_1, z_2^*)$$
 - “Geodesic” $\rightarrow$ shortest path between the 2 points, where the path is restricted to the space (here, hypersphere)
 - Great-circle distance
 - 2) OR approximated by **Euclidean (chord)** distance:
$$\text{distance}(z_1, z_2^*) := \|z_1 - z_2^*\|_2 \text{ (leads to Procrustes distance)}$$



Shape Alignment

- Procrustes analysis for shape alignment

1) Standardize location (putting pointset in pre-shape space: part 1 of 2)

- Compute centroid of each pointset
- Subtract centroid coordinates from each point coordinate

2) Standardize scale (putting pointset in pre-shape space: part 2 of 2)

- Re-scale each pointset to have same scale/“size”
- Size = norm of vector (= some fixed pre-decided number)

3) Align w.r.t. rotation

- How ? We'll see next

4) Procrustes dissimilarity/‘distance’

- = Euclidean distance between optimally-transformed shapes
- Actually, ‘distance’ metric only when pointsets lie in shape space

Shape Alignment

- Procrustes
- Greek meaning: “the stretcher [who hammers out the metal]”
- Rogue, bandit
- Had an infamous bed
- Invited travelers to spend the night in his home
- Had to “fit” guests to the bed
- If the guest proved too short, ...
- If the guest proved too tall, ...
- End: Theseus captured Procrustes, compelling him to be “fitted” to his own bed



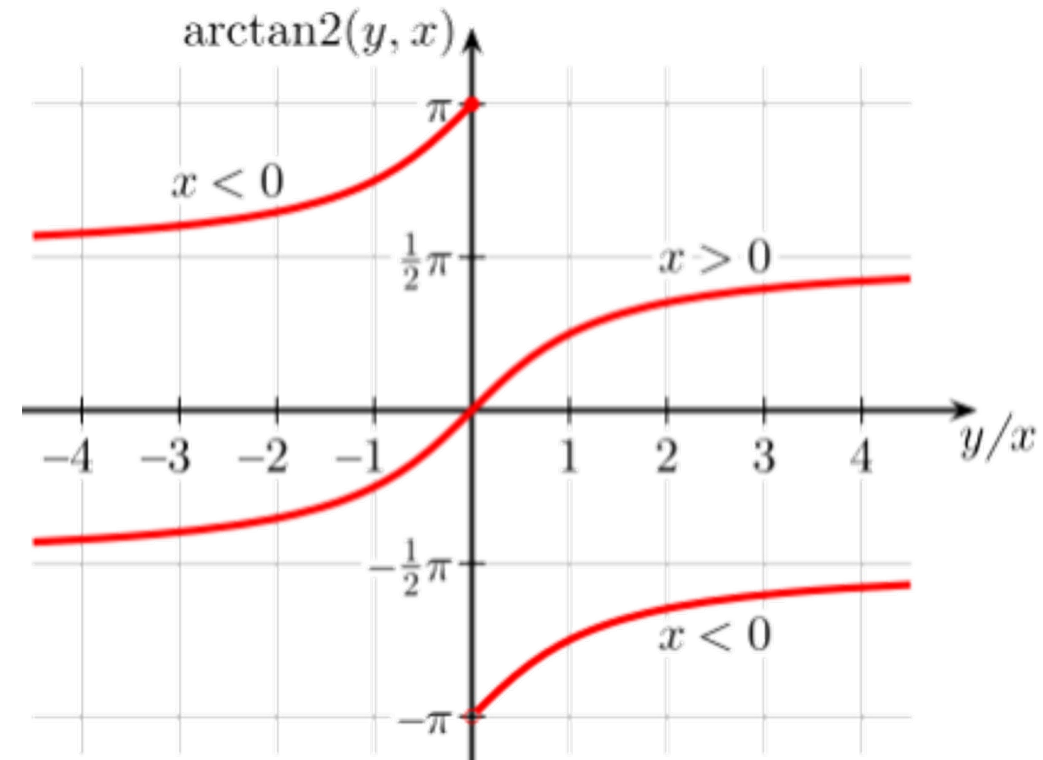
Shape Alignment

- Procrustes analysis for shape alignment (take 2)
- A slightly different take by: [Timothy Cootes et al. 1995. Active Shape Models – Their Training and Application. Computer Vision & Image Understanding.] that solved for all:
 - 1) Translation: 2 variables
 - 2) Scale: 1 variable
 - 3) Rotation: 1 variable
- Assuming z_1, z_2 don't lie in preshape space
- Procrustes dissimilarity between z_1, z_2 (by aligning z_2 to z_1) is
= $\min_{\theta, T, s} d^2(z_1, \text{SimilarityTransform}(z_2; \theta, T, s))$

Shape Alignment

$$M_{\theta} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

- Procrustes analysis for shape alignment (take 2)
- Cootes et al. formulation: $\min_{\theta, T, s} \sum_{n=1, \dots, N} \| z_{1n} - s M_{\theta} z_{2n} - T \|_2^2$
- First substitute $a := s \cos \theta$, $b := s \sin \theta$ and solve for $\{ t_x, t_y, a, b \}$. How?
 - Quadratic in $\{ t_x, t_y, a, b \} \rightarrow$ convex
 - Closed-form solution by solving a **linear system of equations**
 - doi.org/10.1006/cviu.1995.1004
- Then, solve for $s > 0$ and θ , given a and b
 - $s = \sqrt{a^2 + b^2}$
 - So, $s \geq |a|$ and $s \geq |b|$
 - So, $|\cos \theta| = |a/s| \leq 1$, $|\sin \theta| = |b/s| \leq 1$
 - $\theta = \arctan2(b, a)$



Shape Alignment

- Procrustes analysis for shape alignment (take 2)
- Cootes et al. formulation: $\min_{\theta, T, s} \sum_{n=1, \dots, N} \| z_{1n} - s M_{\theta} z_{2n} - T \|_2^2$
- **Optimal translation** T turns out to be ... ?
 - For any given s, θ :
optimal T = displacement between centroids of z_1 and transformed z_2
 - Begin with both centroids at origin because,
once centroid is at origin, changing scale and rotation doesn't change centroid
- **Optimal scale** s is slightly different !
 - Previously (when z_1, z_2 were in preshape space),
rotated pointset z_2 was ensured to be unit norm
 - If unconstrained minimization over “ s ”, that guarantee is lost !

Shape Alignment

- Optimizing for rotation (in $D > 1$ dimensions)
- Given: 'n' points in 2 pointsets $\{ \mathbf{x}_i : i=1, \dots, n \}$ and $\{ \mathbf{y}_i : i=1, \dots, n \}$
- Goal: Find rotation R to best align pointsets

$$\operatorname{argmin}_R \sum_{i=1}^n w_i \|R\mathbf{x}_i - \mathbf{y}_i\|^2$$

- assume $w_i > 0$

$$= \operatorname{argmin}_R \sum_{i=1}^n w_i (\mathbf{x}_i^T \mathbf{x}_i - 2\mathbf{y}_i^T R\mathbf{x}_i + \mathbf{y}_i^T \mathbf{y}_i) \quad \text{Because } R^T R = \text{Identity}$$

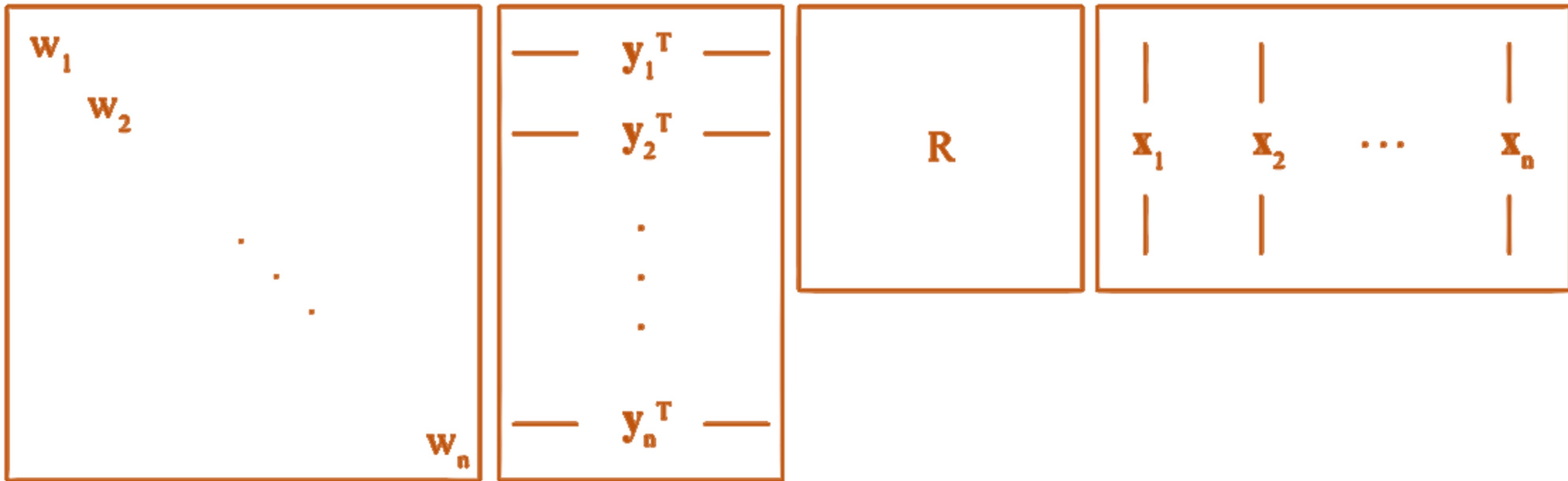
$$= \operatorname{argmax}_R \sum_{i=1}^n w_i \mathbf{y}_i^T R\mathbf{x}_i$$

- under constraint that R is a rotation matrix

Shape Alignment

- Optimizing for rotation (in $D > 1$ dimensions)
- [Kabsch 1976] en.wikipedia.org/wiki/Kabsch_algorithm

$$\sum_{i=1}^n w_i \mathbf{y}_i^T R \mathbf{x}_i = \text{tr} (W Y^T R X)$$



Shape Alignment

- Optimizing for rotation (in $D > 1$ dimensions)
- $\text{Trace}(AB) = \text{Trace}(BA) \rightarrow \text{tr}(WY^T R X) = \text{tr}((WY^T)(RX)) = \text{tr}(RXWY^T)$
- $XWY^T \propto$ (weighted) “cross covariance” matrix
 - Consider SVD of $XWY^T := U\Sigma V^T$

$$= \text{tr}(RU\Sigma V^T) = \text{tr}(\Sigma V^T RU)$$
- Let $V^T R U \rightarrow M$

$$(\Sigma M) = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_d \end{pmatrix} \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1d} \\ m_{21} & m_{22} & \dots & m_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ m_{d1} & m_{d2} & \dots & m_{dd} \end{pmatrix}$$
- Then
- V, R, U = orthogonal matrices $\rightarrow M$ = orthogonal matrix
 $\rightarrow$ elements of M have magnitude ≤ 1

$$= \sum_{i=1}^d \sigma_i m_{ii} \leq \sum_{i=1}^d \sigma_i$$
- How to maximize this quantity, over all orthogonal M ?
 - Set $m_{ii} = 1$, other elements = 0
 - Is M orthogonal ? Yes. So, R is orthogonal
 - Thus, $I = M = V^T R U \Rightarrow V = R U \Rightarrow R = V U^T$

Shape Alignment

$$\sum_{i=1}^d \sigma_i m_{ii} \leq \sum_{i=1}^d \sigma_i$$

- Optimizing for rotation (in $D > 1$ dimensions)
- Are we done ?
- We are **done** if $R = VU^T$ a rotation matrix, i.e., if $\det(R) = \det(VU^T) = +1$
- If $\det(VU^T) = -1$, then
[Umeyama 1991 TPAMI] [Kanatani 1994 TPAMI] proved that best solution has objective function = $\text{tr}(\Sigma M) = \sigma_1 + \sigma_2 + \dots + \sigma_{d-1} - \sigma_d$
- Then $M = V^T R U = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 & \\ & & & & -1 \end{pmatrix} \Rightarrow R = V \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 & \\ & & & & -1 \end{pmatrix} U^T$
- Is M orthogonal now ? Yes. So, R is orthogonal.
- Is $\det(M)$ negated ? Yes. So, $\det(R)$ also negated.

Statistical Shape Analysis

Hermann Karcher

- What is the **mean shape** ?
 - Cannot simply average (across pointsets) coordinates of corresponding points. Why ?
- 1) Mean(s) of dataset =
“center(s) of mass” of dataset =
minimizer(s) of sum of squared distances to data, given some distance metric
 - Karcher mean (set): set of all minimizers
 - If Karcher mean is unique $\rightarrow$ called Frechet mean
 - 2) Mean as maximum-likelihood estimate
 - Assume isotropic-Gaussian PDF on shapes
 - Distance between shapes = Procrustes distance



Statistical Shape Analysis

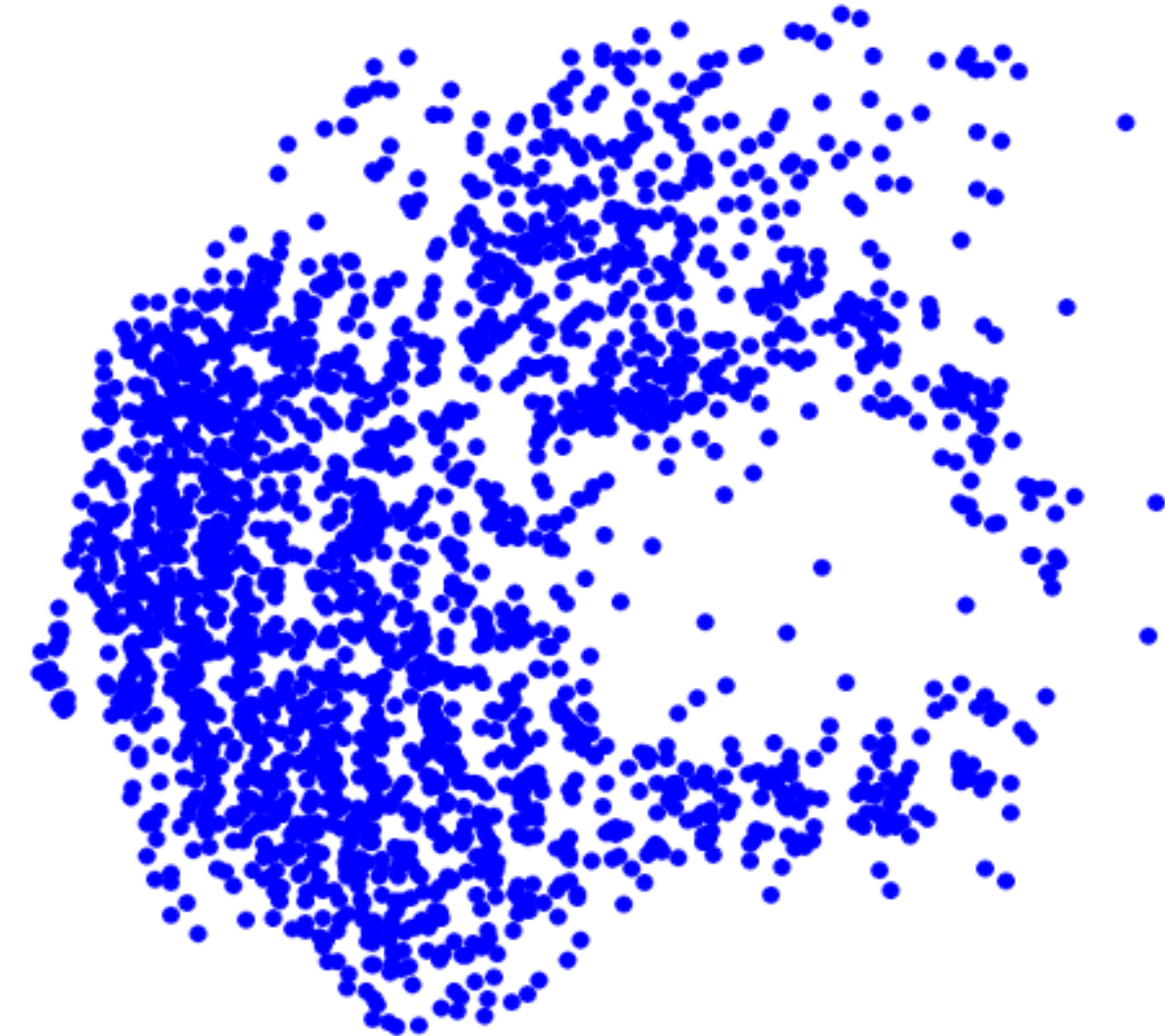
- Mean shape :=
pointset (in preshape space, i.e., with centroid at origin & unit norm)
minimizing sum of squared distances to all given pointsets (preshapes)
- Data $\{ \mathbf{z}^m := m\text{-th observed pointset out of } M \text{ pointsets} \}$
Each $\mathbf{z}^m$ has N points.
Each $\mathbf{z}^m$ is pre-processed to have centroid at origin & unit norm.
- $\mathbf{z} := \text{mean pointset}$ (unknown) also of N points
Constraints: centroid at origin, fixed norm (say, 1; else trivial solution)
- $\{ \mathbf{s}^m, \mathbf{R}^m, \mathbf{T}^m \} :=$
similarity-transform parameters for each data pointset (unknown)
- Minimize, over $\mathbf{z}$ and $\{ \mathbf{s}^m, \mathbf{R}^m, \mathbf{T}^m \}$,
 $\sum_{m=1, \dots, M} \sum_{n=1, \dots, N} \| \mathbf{z}_n - \mathbf{s}^m \mathbf{R}^m \mathbf{z}_n^m - \mathbf{T}^m \|_2^2$ subject to constraints on $\mathbf{z}$

Statistical Shape Analysis

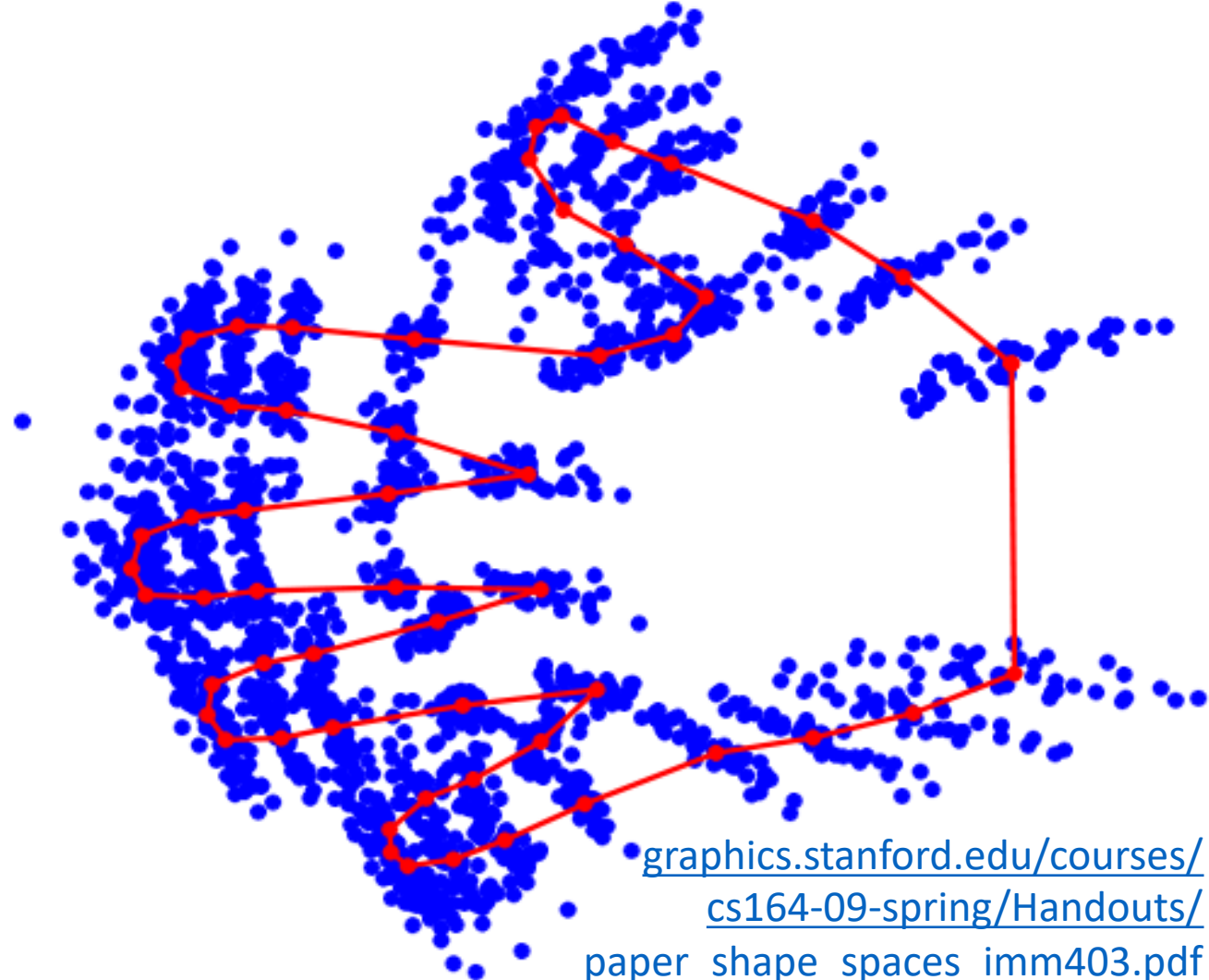
- Minimize, over $\mathbf{z}$ and $\{ \mathbf{s}^m, \mathbf{R}^m, \mathbf{T}^m \}$,
 $\sum_{m=1, \dots, M} \sum_{n=1, \dots, N} \| \mathbf{z}_n - \mathbf{s}^m \mathbf{R}^m \mathbf{z}_n^m - \mathbf{T}^m \|_2^2$ subject to constraints on $\mathbf{z}$
- How to optimize ? Alternating minimization
 - 1) Given mean $\mathbf{z}$, find optimal transformations $\{ \mathbf{s}^m, \mathbf{R}^m, \mathbf{T}^m \}$
 - Solve independently for each $\mathbf{z}_n^m$; if pointsets in preshape space, $\mathbf{T}^m$ not needed
 - 2) Given all transformations $\{ \mathbf{s}^m, \mathbf{R}^m, \mathbf{T}^m \}$, find optimal mean pointset $\mathbf{z}$
 - a) $\mathbf{z}' =$ average all (aligned) pointsets $\mathbf{z}^m'$ where $\mathbf{z}_n^{m'} := \mathbf{s}^m \mathbf{R}^m \mathbf{z}_n^m - \mathbf{T}^m$
 - Similar to gradient-descent step
 - Resulting pointset guaranteed to have centroid at origin. Why ?
Centroid constraint automatically satisfied
 - b) Optimal $\mathbf{z} =$ Take $\mathbf{z}'$ and rescale (divide) by its norm
 - Projection onto constraint set (following gradient step)

Statistical Shape Analysis

- What is the mean shape ?
- Left: unaligned pointsets.



Right: aligned pointsets + mean shape.

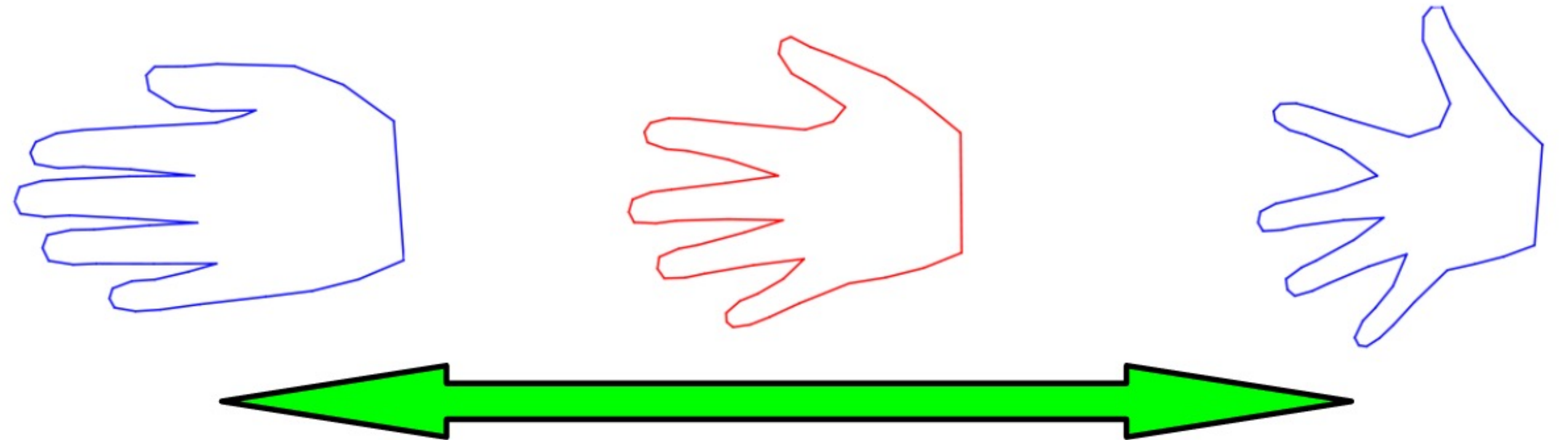
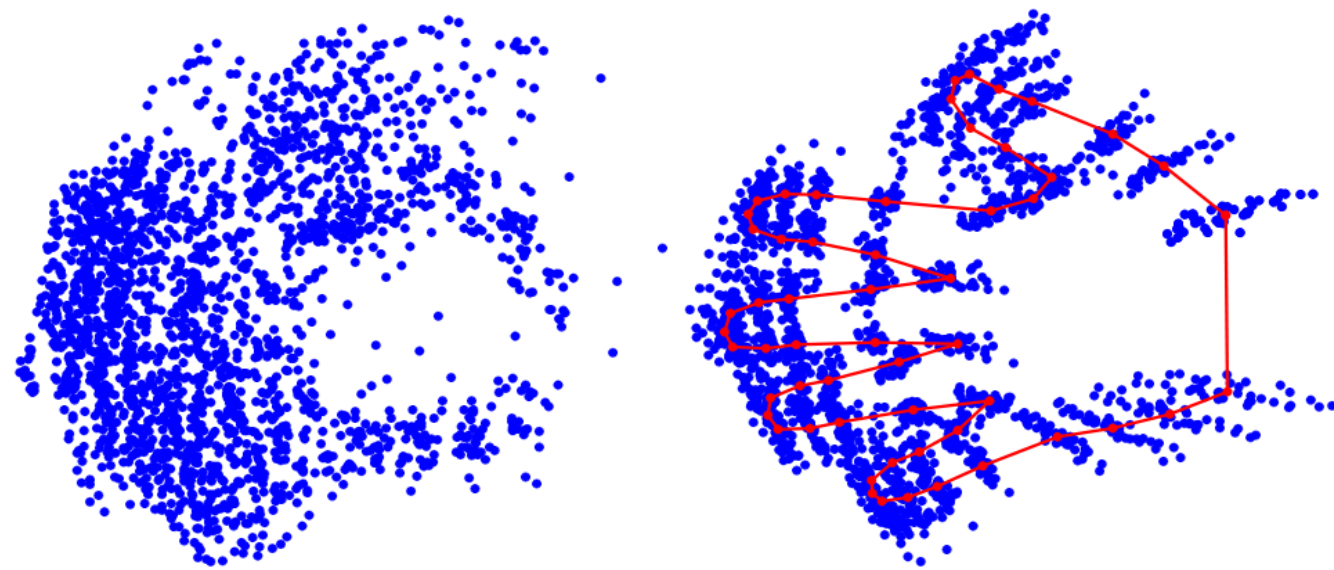


Statistical Shape Analysis

- How to learn **shape variability (around mean)** ?
- Assume:
shape mean is given; all pointsets have been aligned to the mean
 - Optimized using previous approach
- Estimate a shape covariance matrix
 - Maximum-likelihood estimation,
assuming a Gaussian model with a covariance matrix
 - Sample covariance
- Principal eigenvectors of covariance matrix
give principal modes of variation

Statistical Shape Analysis

- Hand dataset
- First eigenvector



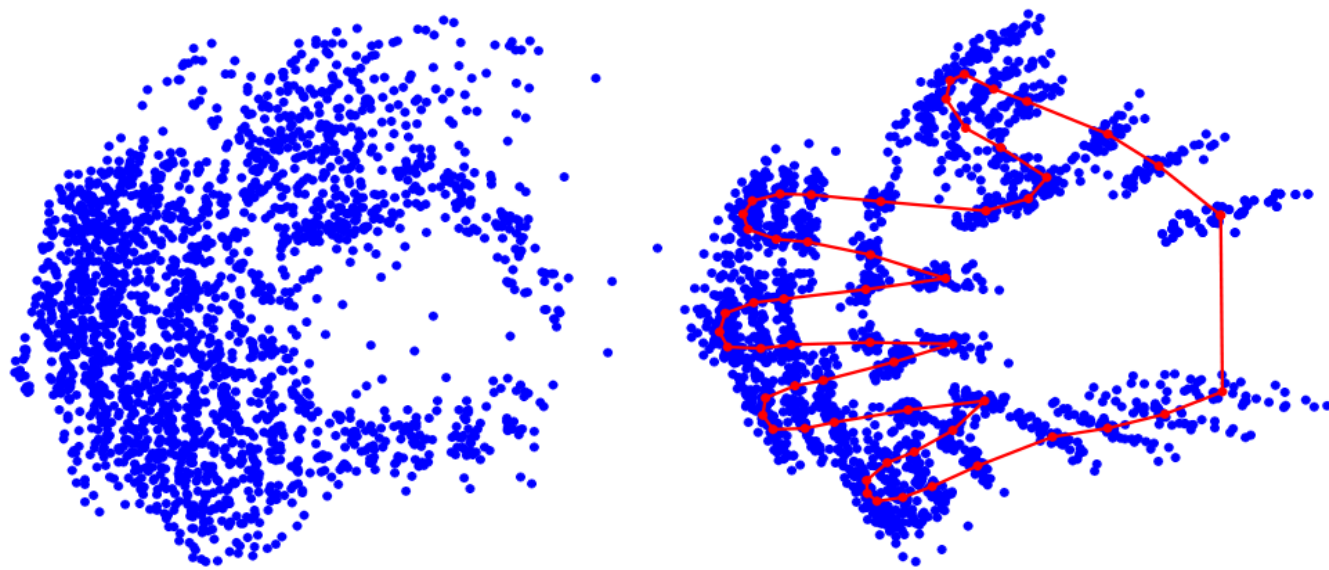
(a) $b_1 = -3\sqrt{\lambda_1}$

(b) $b_1 = 0$

(c) $b_1 = +3\sqrt{\lambda_1}$

Statistical Shape Analysis

- Hand dataset
- Second eigenvector



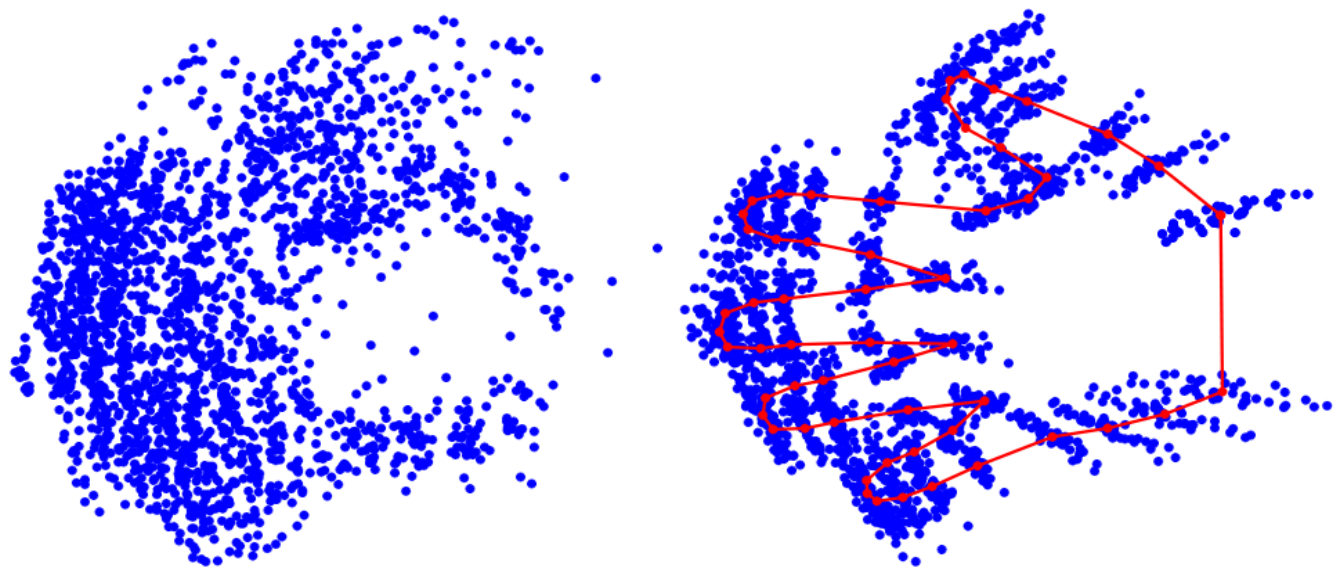
(d) $b_2 = -3\sqrt{\lambda_2}$

(e) $b_2 = 0$

(f) $b_2 = +3\sqrt{\lambda_2}$

Statistical Shape Analysis

- Hand dataset
- Third eigenvector



(g) $b_3 = -3\sqrt{\lambda_3}$

(h) $b_3 = 0$

(i) $b_3 = +3\sqrt{\lambda_3}$

Statistical Shape Analysis

- Bone dataset: principal modes of variation

Mode	1	2	3	4
(-3SD, Mean Shape, +3SD)	 (a)	 (b)	 (c)	 (d)
Percentage of Variance (%)	31.9	19.0	10.6	5.0
Description	Overall length (T_1)	Sternal diameter (T_2) and acromial diameter (T_3)	Medial curve radius (R_1) and inferior curve radius (R_2)	Overall diameter (T_4)

* The green, blue and red models represent the -3 SD, the median and the +3SD models

Statistical Shape Analysis

- Knee dataset: principal modes of variation

Mode 1



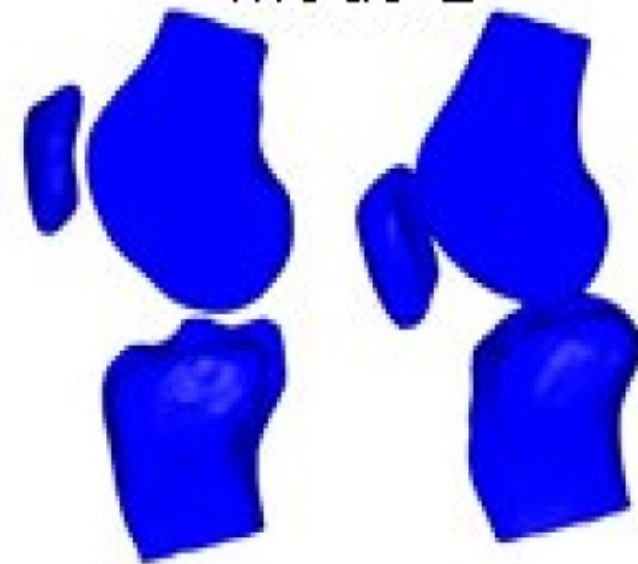
Mode 2



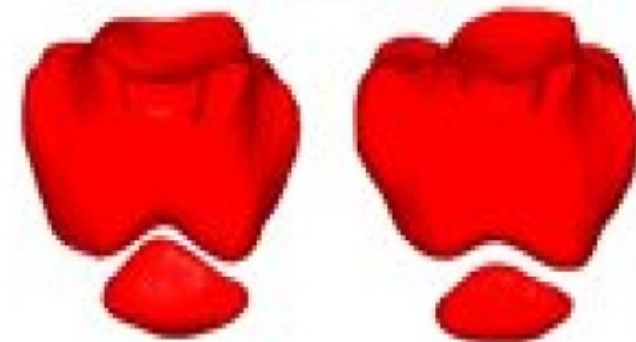
Mode 3



Mode 2



Mode 3



Statistical Shape Analysis

- Medical applications of learning statistical shape models
 - 1) Scientific study
to understand variability
 - 2) Hypothesis testing
to test for shape differences in a specific disorder/disease
 - 3) Classification
of a patient based on shape
 - e.g., shapes of brain structures, shapes of bones
 - 4) Shape priors
for segmentation

Statistical Shape Analysis

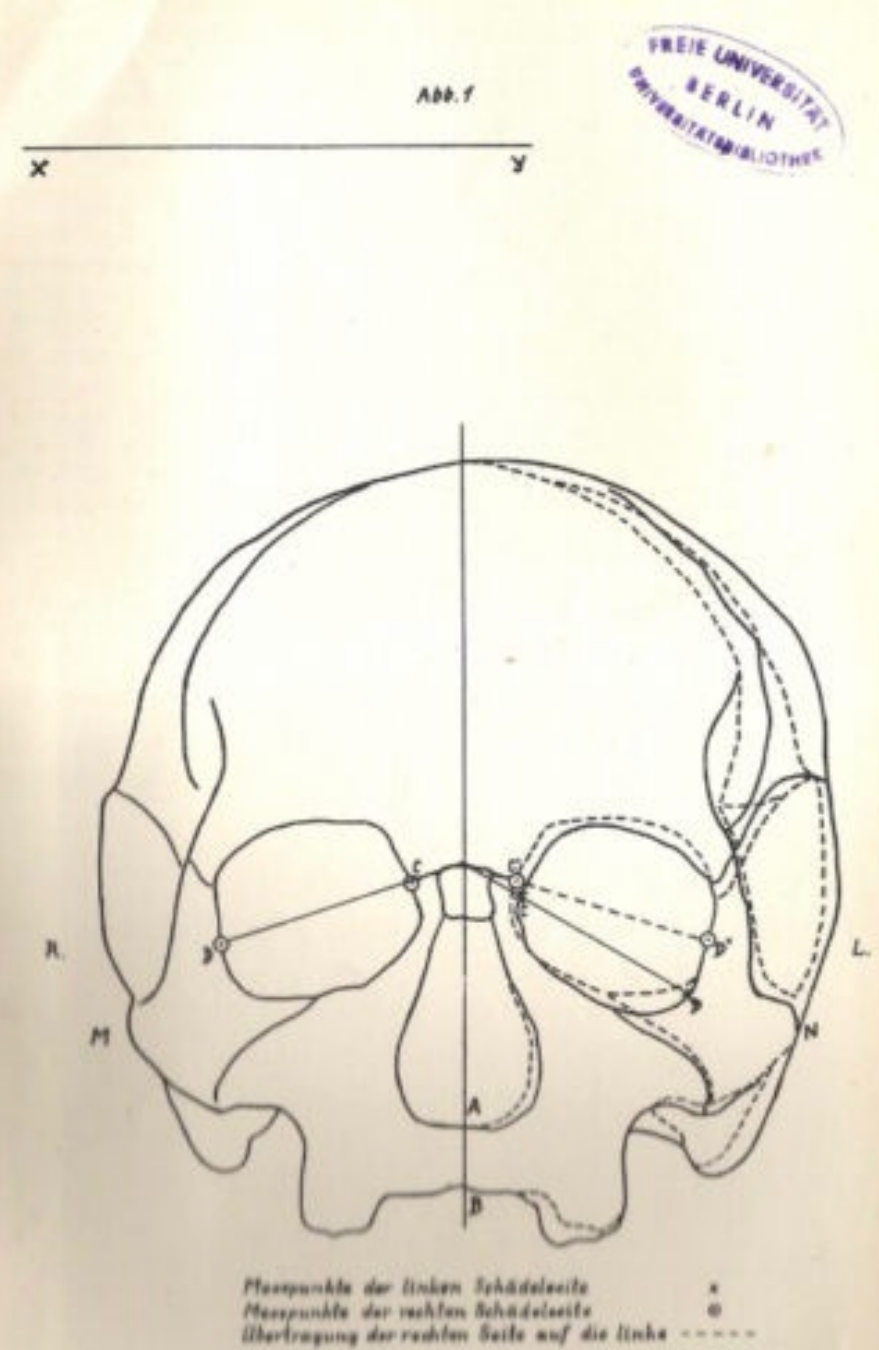
- First Indian woman anthropologist: [Irawati Karve](#)
- PhD: “The normal asymmetry of the human skull”
 - Empirical study involved 149 human skulls
 - In 1928. Age 22. In Germany. [[profile](#)]
- Guide: Eugen Fischer, German anthropologist
 - www.bbc.com/news/articles/c93qqq5g416o
 - **Hypothesis**: white Europeans had **asymmetrical skulls** to accommodate **larger right frontal lobes**, supposedly a marker of **higher intelligence** (and so they were racially superior to non-white Europeans)
 - Irawati found no correlation. Fischer passed her with lowest grade. Later, Fischer joined Nazis.



Statistical



[migrantknowledge.org/
2021/11/09/contradicti
ons-of-irawati-karve](https://migrantknowledge.org/2021/11/09/contradictions-of-irawati-karve)



Statistical Shape Analysis

- First woman in Pune to ride a scooter
- Irawati with her Lambretta scooter, 1950

